

**SKY SHIELD: A SMART DRONE SWARM FOR
REAL TIME UAV IDENTIFICATION AND
THREAT MITIGATION**

A PROJECT REPORT

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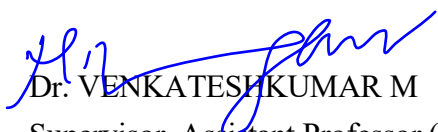


BONAFIDE CERTIFICATE

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ABSTRACT

Due to the fast development of the use of Unmanned Aerial Vehicles (UAVs), the safety of the airspace has encountered various challenges. This has been experienced in areas such as the military area, airfields, and human beings' gatherings. The existing anti-UAV technology has not been effective in the past due to its inability to adapt rapidly to the situation.

The project titled "SkyShield: A Smart Drone Swarm for Real-Time UAV Identification and Threat Mitigation" has developed an autonomous technology for multiple UAVs. The technology uses the concept of radar technology for the detection of the UAVs. The validation of the UAVs' identities has been done using blockchain technology.

The technology has been effective in the development of a scalable, smart, and safe technology for the protection of the UAV airspace. This is evident by the working prototype of the technology, which has been developed. The technology uses the concept of blockchain technology, YOLOv8, and radar simulation.

The implemented technology has provided a scalable, smart, and secure technology for protecting UAV airspace, as shown by the working prototype of the technology, which comprises blockchain technology, YOLOv8, radar simulation, and real-time monitoring.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	EXPANSION
1	ACO	Ant Colony Optimization
2	AES	Advanced Encryption Standard
3	AI	Artificial Intelligence
4	API	Application Programming Interface
5	CNN	Convolutional Neural Network
6	DAGSVM	Directed Acyclic Graph Support Vector Machine
7	DNN	Deep Neural Network
8	FANET	Flying Ad Hoc Network
9	FSS	Fish School Search
10	GANACHE	GUI-based Blockchain Simulator for Ethereum
11	MJPEG	Motion JPEG (Joint Photographic Experts Group)
12	GUI	Graphical User Interface
13	HTTP	HyperText Transfer Protocol
14	ID	Identification
15	IoT	Internet of Things
16	LSTM	Long Short-Term Memory
17	ML	Machine Learning
18	GPS	Global Positioning System
19	mAP	Mean Average Precision
20	MAVLink	Micro Air Vehicle Link
21	PSO	Particle Swarm Optimization
22	RF	Radio Frequency
23	RL	Reinforcement Learning
24	RSSI	Received Signal Strength Indicator
25	SI	Swarm Intelligence
26	SNR	Signal-to-Noise Ratio
27	UDP	User Datagram Protocol
28	UAV	Unmanned Aerial Vehicle
29	VMD	Variational Mode Decomposition
30	YOLO	You Only Look Once

Chapter 1

INTRODUCTION

1.1 PREFACE

Unmanned Aerial Vehicles have been employed in other fields such as defense mechanisms, transport mechanisms, observation systems in agriculture, monitoring systems, and inspection systems in industry. Despite the many advantages that come with the use of UAVs, there are also threats to security that have arisen as a result of the misuse and unauthorized use of these systems.

The current anti-drone systems that make use of radar sensing, radio frequency detection, and visual detection are faced with the problem of scalability and secure identification. This problem is even more pronounced in urban areas, low-altitude areas, or areas that are experiencing conflict. The challenge of identifying friendly, commercial, and enemy drones is a major problem in these areas.

There is therefore a need for the development of intelligent and secure multi-drone defense systems that can detect, classify, and respond to aerial threats in real time. In addition, recent studies have indicated the possibility of using a combination of swarm intelligence and blockchain technology in order to ensure a decentralized decision-making process. In this case, the communication process would be tamper-proof, leading to a better trust level in the cooperation of drones in the management of aerial threats.

1.2 OBJECTIVE

The fundamental purpose of this project's initiative is to create and implement an independent unmanned aerial vehicle (UAV) multi-UAV network capable of real-time detection, identification, and response to unauthorized drone activity. This new way should add intelligence to airspace security, while at the same time applying fast threat analysis methods to the integrated network of UAVs being deployed in the airspace. To achieve this goal, the developed UAV system integrates both a radar scanning and encrypted GPS signal confirmation dual mode detection method. Combined, these methods will create an autonomous UAV network that has more situational awareness regarding its surrounding environment, and thus will enable improved accuracy and reliability in tracking/monitoring targeted drones throughout their entire flight.

The use of Blockchain for authenticating and verifying drone credentials will provide the necessary security and immutability for managing drone identity. This eliminates the risk of single point failures or unauthorized access, data alteration, or spoofing. The project will create an immutable record of drone identity and activity for promoting accountability and trust amongst UAVs.

The project is dedicated to protecting airspace through ethical and non-destructive counter-measures. The project will use warning and tracking methods in lieu of intercepting. There will also be a Ground Control Station with an interactive graphical user interface for real-time monitoring and effective communication between the UAVs' operators and the human operators. This project will facilitate proper coordination and decision-making between human operators and autonomous UAVs (unmanned aerial vehicles) units.

1.3 ORGANISATION OF THE REPORT

Chapter 1 - Introduction:

This chapter presents an overview of the evolving threat landscape associated with UAVs and develops a rationale for developing a unified and intelligent defense framework. The problem of unauthorized UAV activities is defined, and the objectives of the Sky Shield system proposed later are outlined. The chapter provides the necessary background context to outline the challenges related to the detection and mitigation of illegal UAV operations.

Chapter 2 - Literature Review:

This chapter provides a comprehensive review of the existing literature in the related areas of swarm intelligence, methods of detecting unauthorized UAVs, AI-based payload and threat analysis, secure communication mechanisms, and blockchain technology deployment in UAV networks. Identification of key research gaps in the current state-of-the-art approaches is also discussed here, which had been considered for the design and development of the proposed SkyShield framework.

Chapter 3 - Methodology:

This section presents the principles and methodology of implementation of the proposed Sky Shield system. It outlines the overall architecture of this system, its operational workflow, and the technologies used. The functions of the various system components are discussed, together with the expected outcomes and current progress made in the development and prototype implementation of the system.

Chapter 4 - References:

This chapter provides all the academic and technical references used in the entire report. These include journals, conference papers, and all other credible sources that were used in the development of this document.

1.4 BACKGROUND AND NEED FOR UAV SECURITY SYSTEMS

The rapid evolution of technology in the area of unmanned aerial systems (UAS) has fundamentally changed numerous industries such as defense; agriculture; logistics; and observation tools. The adoption of UAS by both commercial and civilian users combined with the many benefits of using a UAS (i.e., flexible; economical; and accessible) has led to the large-scale, widespread usage of UAS. As a consequence, this widespread use of UAS has created new challenges in terms of how to maintain safe and secure environments for operations conducted in the airspace above us.

The increased volume of low-cost UAS available for use means that the risk associated with the use of UAS for non-authorized purposes has dramatically increased. An UAS that has been used for unauthorized operations can represent an extreme risk to many elements of our society, including public safety; private citizen privacy; and our nation's critical infrastructure. The introduction of even one unidentified UAS within a high security area (e.g., commercial airports; military installations; and large urban areas) will create operational disruption and threaten security efforts. The complexity of the modern airspace (in addition to the fact that UAS operate within our airspace in a dynamic environment) makes it that much harder to effectively monitor and control UAS that operate within our atmosphere.

Conventional UAV detection methods mainly use a combination of various individual technologies: radar, RF and visual detection. Each technology has its perceived strengths but also has significant weaknesses. For example, radar systems can be costly and inefficient in detecting small type drones. RF-based detection systems are very susceptible to interference. Visual detection systems are heavily reliant upon environmental conditions including lighting and weather.

Consequently, when one of the three is relied upon at a minimum, typically leads to incomplete/unreliable detection outcomes.

In addition to detection challenges, secure identification remains another pressing issue. Most technology currently available, fails to provide an adequate means of validating the identity of detected drones, resulting in increased risk of theoretical spoofing and unauthorized access. In addition, many systems use centralized databases for managing drone identity, which makes them vulnerable to cyber threats and manipulation of data; corrects contributing to the reduced confidence in the overall infrastructure.

The next key element is the requirement for autonomous operation within very short, real-time, response timeframe. In many instances where there are delays associated with detection or validation, have the ability to cause catastrophic results. Therefore, developing systems to operate in an autonomous fashion, with very low latency, while maintaining a high degree of reliability and accuracy, will be absolute necessity moving to the future.

As a result of these challenges, there is an increasing requirement for a comprehensive and intelligent counter-UAV defense system capable of using many detection methods, having robust authenticating protocols, and being able to process information quickly. This type of system should also function effectively in a dynamic environment while ensuring that it is scalable, secure, and complies with ethical standards.

1.5 SCOPE OF THE PROJECT

SkyShield System application area includes specific regions for implementing, validating and controlled testing of an integrated UAV defence system (UAV = Unmanned Aerial Vehicle). The prototype of SkyShield has been intentionally designed to operate within indoor environments to ensure that safety, viability and testing results are accurate during the prototyping stage.

The system is developed for short-range indoor prototype testing with two ESP32 microcontrollers acting as an authorized UAV and an unauthorized intruder UAV. The authorized ESP32 microcontroller is equipped with a laser module interfaced with GPIO25 to verify a controlled physical threat response mechanism. The unauthorized ESP32 microcontroller sends a distinct UAV ID to mimic spoofing attacks and unauthorized airspace entry, thereby facilitating the real-time testing of the verification logic of the system.

YOLOv8-based visual detection is developed on a Raspberry Pi 4 with a customized drone detection model. The system analyzes a live video stream to detect the presence of drones and identify them with confidence scores, facilitating real-time decision-making. This module verifies the viability of using deep learning-based detection on edge devices with restricted computational capabilities.

The Ethereum blockchain network running on Ganache is responsible for the management of identity registration and verification through the DroneRegistry smart contract. All requests for drone verification and authorization results are recorded immutably on the blockchain network with distinct transaction hashes. This module proves the secure and decentralized management of identity in a UAV defense system.

The object detection simulator uses HC-SR04 ultrasonic sensors and SG90 servo motors for two-dimensional radar scans or proximity detection within a 200cm area (not an actual radar). Therefore, this simulator meets the requirements of proximity detection and situational awareness to support aircraft with a multi-sensor approach.

The DroneShield Flask interface was created to provide an all-in-one solution for real-time monitoring of the system from one web-based interface. This includes a simulated airspace map, detection notifications, updates to the blockchain, and threat notifications, providing the operator with complete situational awareness (or control).

The prototype does not include outdoor GPS function, actual Neo-6M GPS modules, the use of any public Ethereum networks, coordinated control of swarms of three or more autonomous drones, or HTTPS encrypted communications. Each of these items has been outlined so that the project remains feasible, safe, and within academic constraints for scalability.

1.6 MOTIVATION

The development of the SkyShield program is based on three primary areas of concern that have been identified as part of the early study and evaluation of drone defense systems. Collectively, these three areas reveal significant flaws in the various types of current anti-drone systems and illustrate some opportunities for enhancing the capabilities of existing systems through integrating them with new technologies.

First, the majority of the established anti-drone systems being used today are either prohibitively expensive (compared to their functionality) for use by researchers or commercial users on a limited budget, or are simplistic in their functionality. Most of the systems currently used in the marketplace are primarily designed to detect the presence of drones but lack any

type of reliable means for verifying the identity of the drone or creating secure records of any potential event that occurs in their airspace (i.e. if the drone violates the airspace). This results in an inability for any existing anti-drone system to create tamper-resistant documentation of any violation of a given airspace, and thus creates a significant number of accountability issues in cases where forensic documentation is needed, regulatory requirements exist, or subsequent investigation occurs following an incident or a violation of the airspace.

Two other examples of successful implementation of blockchain technology are in the fields of supply chain management and healthcare and finance. They provide decentralized trust through the use of immutable transaction records. There is no literature investigating how blockchain would be applied to real-time verification of drone identities and airspace security, which is where the SkyShield project fills this gap by using Web3.py and Flask-based blockchain identity verification so that every request for drone identity verification is both verified and recorded.

Third, the YOLOv8 is currently the best-performing algorithm for performing real-time object detection; it has the best combination of speed, accuracy, and resource utilization. When tested at the edge (Raspberry Pi 4) without any reliance on cloud computing services, the YOLOv8 proves that AI systems based on edge computing architecture can perform as good as traditional computing, particularly where there is limited network availability, thus making them high speed and efficient, while increasing the privacy and autonomy in those environments.

This evidence indicates there is a need for an affordable, intelligent and integrated UAV defence system. The SkyShield was developed to be the world's first integrated UAV defence system which incorporates visual detection, blockchain-based authentication, radar modelling, and controlled physical laser response into one complete system. The system aims to bridge the current gap in defence and provide a base for further investigations into autonomous and decentralised aerial defence systems.

1.7 SUMMARY

Unmanned Aerial Vehicles (UAVs) are widely used in military, logistics, agriculture, and industrial applications, but security threats from the misuse of UAVs are significant challenges. The existing anti-UAV systems face scalability, adaptability, and correct identification for security. To overcome the security threats and challenges, this research proposes SkyShield, a highly automated multi-UAV security system that integrates dual-mode detection, secure communication, and Blockchain identification for multi-UAV security applications. This research work has designed an in-depth approach that uses literature analysis and system design concepts to prove the feasibility of intelligence and security for airspace protection in validating the feasibility of intelligence and security for airspace safeguarding.

Chapter 2

LITERATURE REVIEW

2.1 INTRODUCTION

This research aims to highlight the current developments of research in different technical fields to build an efficient and independent defense system against malicious UAVs. It reviews research studies that cover concepts of collaboration and coordination of multiple UAVs, detection methods of UAVs, AI-based analysis methods of payloads, secure communication networks, and the use of blockchain technology in UAV verification processes. The aim of this review is to clarify the current status of recent research developments and identify the research gaps that the proposed SkyShield system tries to fill.

2.2 MULTI-DRONE COLLABORATION AND CONTROL

The two fundamental skills that enable an unmanned aerial vehicle to act autonomously and under complicated setup and circumstances are drone coordination along with swarm intelligence (SI). Swarm intelligence is based on social insect species, including ants and bees, whose simple local interactions create intelligent collective behavior on the global level. Due to self-organizing, decentralized control, UAV swarms can adaptively respond to environmental changes without needing centralized monitoring or control, increasing the swarms' robustness and fault tolerance [1].

A variety of optimization and bio-inspired algorithms have been developed to provide assistance in distributed planning and cooperative control of UAV swarms through the use of Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO) and Fish School Search (FSS), which allow for efficient task allocation, path planning, and formation control under dynamic conditions. Additionally, consensus-based models facilitate the ability to reach collective decisions simultaneously without regard to the level of uncertainty and time delay or the communication limits of the environment [2], [3] have enabled UAVs to agree on shared objectives/targets while maintaining their freedom and independence under decentralized control.

The effectiveness of swarm UAV networks can be seen in a variety of situations, including search and rescue operations, environmental monitoring, resource exploration, and traffic monitoring. When performing these activities, multiple UAVs must cooperate with one another to modify their coverage patterns in real time as well as to share the data they collect and control their energy usage for optimal performance. However, one of the biggest challenges faced by swarm UAVs in these scenarios is their ability to communicate reliably with one another, particularly in ad-hoc and mobile network environments. Research into wireless mesh networks, decentralised control systems, and energy-efficient communication protocols has contributed to improving the robustness of communication links, reducing latency, and enhancing the scalability of swarm UAV systems [3].

Another important area in autonomous swarm UAV deployment is path planning and obstacle avoidance. Research by Zhou et al. [6] shows that constraint-aware and evolutionary-computing-based path planning techniques allow a group of UAVs to move through a complex environment while avoiding collisions with other UAVs. Energy optimization and load balancing techniques have also been investigated to enhance swarm performance during long-duration missions.

In conclusion, swarm intelligence is a key technology in the design of fault-tolerant, scalable, and adaptive multi-UAV systems. The results of previous research work have confirmed that the theoretical and practical basis of cooperative UAV defense systems is very relevant to next-generation autonomous and distributed aerial security systems such as SkyShield.

2.3 UNAUTHORIZED DRONE DETECTION TECHNIQUES

The detection of rogue UAVs has become increasingly important due to the large increase in the misuse of UAVs for purposes like illegal surveillance, smuggling and/or attacking. Many advances have been made in the use of RF (radio frequency) fingerprinting as a primary means of detecting UAVs. This method is based on imperfections in the hardware of the UAV's RF signals at the time of their transmission. Using these RF signal patterns, several deep learning models have been developed for identifying & classifying rogue UAVs, including (but not limited to) CNNs (Convolutional Neural Networks), DNN (Deep Neural Networks) and LSTM (Long Short-term Memory) networks. LSTM-based models have been shown to provide high accuracy levels when detecting rogue UAVs in laboratory conditions, based on capturing the temporal changes of the RF signals.

In addition to LSTM-based models for UAV detection, many recently introduced machine learning-based models have been created to enhance detection in noisy and interference-heavy environments. For example, Composite Ensemble Learning Networks (CEL) utilize wavelet-based de-noising methods together with neural-based classifiers to enhance the quality of the RF signals prior to their classification, therefore enhancing the reliability of UAV detection in poor detection environments [5]. Other examples of improving classification precision levels in noisy, high distortion, and frequency-overlapping environments include the use of VMD (Variational Mode Decomposition) and DAG-SVM (Directed Acyclic Graph Support Vector Machines) methods [7]. Spectrogram-based techniques like the proposed model, namely SMNet, have also shown promising results in analyzing drone control signals in environments where electromagnetic interference is present [8].

However, RF-based detection systems still face challenges in the form of Wi-Fi and Bluetooth interference, multipath fading, and low SNR. To resolve these challenges, adaptive filter methods and contrastive learning methods have been incorporated into RF-based detection systems for better discrimination and robustness of the model. The experimental results presented in the literature show that the RF-based detection system is capable of achieving an accuracy of more than 99% and a classification accuracy of more than 93%. However, the reliance of RF-based detection systems on the quality of the signals and the possibility of congestion in the frequency spectrum are limitations for these systems in urban areas.

As an alternative and complementary solution, vision-based detection using deep learning has been proposed as a feasible and deployable solution. YOLO, which stands for You Only Look Once, is a deep learning object detection tool for detecting objects from video feeds, which can be applied for UAV detection using video feeds from the scene without using RF signals. Yilmaz and Kutbay [13] showed that YOLOv8 has detection accuracy higher than 90% and that it outperforms previous versions of YOLO in terms of both detection speed and accuracy. Vashisht et al. [14] performed a comparative study of different versions of YOLO and showed that YOLOv8 performs best for real-time deployment scenarios. Moreover, Selvi et al. [15] tackled the issue of differentiating drones from birds using deep learning classifiers for distinguishing drones from birds in open-sky environments.

These results, in total, prove that YOLOv8 is the best-suited model for the SkyShield visual detection module, especially for edge-based and real-time use cases. A comparative study of the most frequently used UAV detection methods, their performance evaluation parameters, operating conditions, and their limitations is provided in Table 2.1, showing the need for a multi-modal and integrated detection system for UAVs.

Table 2.1 Comparative Performance of Various UAV Detection Methods.

Method	Key Features	Accuracy/ Performance	Limitations
Acoustic	Sound-based; effective in quiet zone	Not specified (depends on microphone sensitivity and environment)	Affected by ambient noise
RF	DOA estimation, fingerprinting, AI-based models	Up to 98.9% at 10 dB SNR; 93% dataset accuracy	RF interference from other devices
Visual	Camera-based under clear visibility	Not specified	Sensitive to light and weather
ML (RF - based)	LSTM , CNN	95-99% Accuracy	Requires large labeled RF Datasets
ML (Vision -based)	YOLO, SqueezeNet, sequence models	Accuracy > 90%	Perfrmance drops in low light scenes
DL (RF- based)	Transformers, self- attention	87.94% with augmentation	Sensitive to Environment
TinyML	Contrastive learning, edge deployable	+14.16% accuracy boost	Hardware limitations

2.4 AI FOR PAYLOAD AND THREAT ANALYSIS

UAVs also require secure communication with other UAVs, as UAV networks are more prone to cyber attacks, interceptions, spoofing, and unauthorized access. However, considering the limited computing power of UAVs, heavy cryptographic algorithms cannot be employed for ensuring security in UAV communication networks. To address the issue of limited computing power, various lightweight cryptographic algorithms have been developed for ensuring security in UAV communication networks.

Blockchain technology-based authentication mechanisms with the help of Elliptic Curve Cryptography (ECC) have been developed to provide secure communication in UAV networks, ensuring the integrity of the communication process. The improved version of the MAVLink protocol, which incorporates ChaCha20 encryption with Diffie Hellman key exchange, has also been developed to enhance the security of communication in UAV networks against eavesdropping, spoofing, and replay attacks [13].

In addition to these conventional cryptographic methods, recent research has also attempted to introduce advanced security paradigms such as homomorphic encryption and lattice-based quantum cryptography for UAV network security. These methods are expected to offer long-term security against quantum computing attacks in the near future while providing secure data processing capabilities without revealing any plain text information. These solutions, despite their higher computational complexities, demonstrate the increasing focus on secure communication paradigms.

Overall, these solutions demonstrate a paradigm shift towards secure, decentralized, and scalable communication architectures for autonomous UAV operations in hostile environments. The integration of lightweight cryptography with blockchain-based authentication solutions demonstrates a close match with the communication security requirements of next-generation UAV defense solutions such as the proposed SkyShield framework.

2.5 SECURE COMMUNICATION AND ENCRYPTION IN UAV NETWORKS

Secure communication between UAVs is critical, considering the increasing possibility of cyberattacks, interceptions, spoofing, and unauthorized access. The UAVs, being highly constrained in terms of power, memory, and computing power, do not allow the use of traditional cryptographic techniques for communication. To address the need for ensuring the integrity of communication, several lightweight encryption techniques have been developed to provide confidentiality, integrity, and authenticity in communication.

The combination of blockchain based authentication protocols and elliptic curve cryptography will provide extremely secure methods for exchanging secure keys, for decentralised identity verification and for securing the immutable, verified integrity of communications. MAVLink communication protocols have also been enhanced by using ChaCha20 for encryption, and Diffie-Hellman for key exchange which greatly reduce the ability of an eavesdropper to see a communication and reduce the potential for replay attacks or unauthorised access to communications, while ensuring high communication efficiency [13].

In addition to these encryption methods, it is important to incorporate additional security techniques such as homomorphic encryption and lattice based quantum resistant Cryptography to protect against any possible computational threats in the future to UAV communication systems. Even though processing system complexity will likely increase with these new technologies, the levels of assurance provided by these methods are currently at an extremely high level when it comes to providing a safe means of transmitting data. In general, these systems indicate an increased focus on creating scalable, real-time, and robust communication systems for UAVs.

2.6 BLOCKCHAIN FOR DRONE VERIFICATION

UAS-related information can be dependent upon the reliability provided through the decentralized and immutable attributes that blockchain technology offers by securing the authenticity, traceability and integrity of communications. With the implementation of a blockchain system, UAS-related information would not permit unauthorized modification or deletion [14]. This could be very important for UAV missions where security is critical.

An authentication protocol that utilizes lightweight blockchain technology and provides secure communication between UAVs and ground stations has been developed through an efficient key management scheme (low-latency) [15]. The design of this protocol removes the need for a centralized authentication server, diminishing the chances for spoofing and replay attacks. use of blockchain technology in Flying Ad Hoc Networks (FANET) enables the implementation of decentralized identity verification for multiple UAVs, thus ensuring secure communication and resistance to various cyberattacks.

Despite all these advantages, there are challenges related to the computational and storage overheads involved in implementing blockchain technology, which might be an issue for UAVs with limited resources. Therefore, to overcome all these challenges and limitations, recent research is focused on developing lightweight blockchain models and storage solutions, and permissioned blockchain models, which might be useful in implementing blockchain technology for developing an UAV verification system such as SkyShield.

2.7 RESEARCH GAP

In spite of the major breakthroughs achieved in drone detection, identification, and mitigation technologies, some major research gaps have not been addressed. The first major gap is the lack of a unified and integrated framework, with most technologies being designed to only provide a single feature, such as drone detection, identification, or mitigation, rather than integrating all the technologies together to provide a complete security solution.

Although the use of blockchain technology has shown promise for ensuring the security of drone identification and tracing, it has not been utilized for real-time drone authentication and ethical decision recording. The lack of seamless integration with drone technologies has created a lack of trust, auditability, and transparency, which is a major concern when it comes to drone security. In addition, most technologies have not been designed with a focus on lightweight blockchain technologies.

The present research in the area of C-MUAVs is still in its infancy, mainly focusing on theoretical or simulation-based approaches rather than deployment strategies in real-world surveillance and counter-UAV scenarios. Similarly, most DL models used in threat classification and detection mechanisms heavily rely on cloud-based processing mechanisms.

The limited use of edge AI in platforms like Raspberry Pi and ESP32 also limits the overall responsiveness and independence of the systems. The use of AI inference mechanisms in edge devices can be very beneficial in decision-making mechanisms.

Moreover, one also observes the lack of research in the area of ethical and non-destructive threat mitigation mechanisms. Most counter-UAV systems use aggressive mechanisms like RF jamming or kinetic intercepts, which are usually destructive in nature and can cause collateral damage to the surrounding area.

Lastly, the overall GUI design is also not properly addressed in most counter-UAV systems, where the overall GUI does not incorporate radar, camera, telemetry, and blockchain-based authentications in real-time scenarios. A unified and interactive GUI can be very beneficial in overall decision-making mechanisms, which is also addressed in the proposed SkyShield system.

Table 2.2 Comparative Analysis of Existing UAV Security Systems

Reference	Detection Method	Authentication	Blockchain	Real-Time GUI	Limitation
Ali et al. [1]	Swarm ACO	None	No	No	No identity verification
Yan et al. [7]	RF + Deep Learning	None	No	No	No visual or GPS module
Yilmaz [13]	YOLOv8	None	No	No	No authentication module
Kumar et al. [17]	None	Blockchain	Yes	No	No detection module
Zaghdoud et al. [25]	None	FANET Blockchain	Yes	No	No visual or radar
SkyShield (This Work)	YOLOv8 + Radar	Blockchain (Ethereum)	Yes	Yes	Indoor prototype only

2.8 COMPARATIVE ANALYSIS OF EXSISTING SYSTEMS

A structured comparison of the significant systems and the research works discussed in this chapter is presented in Table 2.2. The comparative analysis compares the systems based on critical parameters such as detection mechanisms, identification mechanisms, communication security, scalability, and the ability to function in real-time environments. This evaluation will help to understand the advantages and disadvantages of the current state of affairs, thus providing a direct basis to understand the current research gaps.

A comparative analysis shows that the current focus of technology in this field is on particular technology functionalities, such as RF detection; visual classification; communications security, etc., when, in fact, there should be an all-encompassing solution for this entire issue. The current status of technology in the RF area indicates that systems using RF produce a high level of accuracy within a controlled environment; however in chaotic environments, interference caused by devices causes severe difficulties due to spectrum overuse. The current status of technology in the visual classification space indicates that systems using visual classification will produce a good level of accuracy; however, the lack of a secure mechanism for establishing the identity of the person being classified is a large issue that is not solved today.

Furthermore, blockchain-based UAV solutions are still in the conceptual phase and are mostly simulated solutions, with little to no deployment in real-time and minimal integration with detection/visualization tools. Existing solutions are also lacking in comprehensive interfaces for the operators, as most solutions have not successfully combined detection, identity verification, and potential threats in an integrated real-time graphical user interface.

It is clear from the comparison that none of the existing solutions have successfully combined the four features of radar-based sensing, YOLOv8-based visual detection, blockchain-based drone authentication, and real-time GUI visualization into one cohesive solution. The proposed SkyShield solution is unique in that it successfully integrates all four features into one cost-effective and fully integrated UAV defense solution.

2.9 CRITICAL ANALYSIS OF EXISTING UAV DETECTION AND SECURITY METHODS

Current UAV detection and mitigation technologies have made great strides over the last few years, but do have their limitations when deployed in the real world. Many of the current technologies available apply to isolated independent areas of concern related to UAV security, such as UAV Detection, UAV Identification, and UAV Communication, rather than providing a cohesive strategic UAV security framework.

Coordination techniques, using swarm intelligence, have demonstrated highly effective means of decentralized control of multiple UAVs because of their ability to enable distributed decision making across multiple UAVs. For example, when using swarm intelligence coordination algorithms such as ant colony optimization (ACO) and particle swarm optimization (PSO), UAVs are able to efficiently allocate tasks between themselves and adapt to dynamic task environments [1], [2]. The primary focus of these coordination techniques is for UAVs to coordinate their activities with one another; however, they do not integrate real-

time threat detection and authentication features into these types of coordination systems. Additionally, because of the impacts of latency and packet loss in swarm networks, communication issues can have a significant effect on the overall reliability of these UAV systems [3].

Decentralized coordination models and consensus-based control algorithms improve the fault tolerance and scalability of multi-UAV networks [4], [5]; however, these approaches too often lack secure identity verification mechanisms resulting in their susceptibility to spoofing and unauthorized access.

Drone identification methods based on RF technology are proving to be highly accurate for identifying UAVs via signal features. There are methods that use RF fingerprinting which uses small imperfections in the hardware used to make the UAV for identification purposes. In addition, certain advanced methods that use deep learning techniques and modes of signal decomposition (e.g., Variational Mode Decomposition - VMD), have also increased the accuracy of classifying UAVs. Finally, ensemble learning methods can be used to make the UAV detection methods more robust when using RF verification methodologically, especially in noisy environments, such as urban areas.

Spectrogram-based identification methods have also been investigated to identify UAVs, using model types such as SMNet to identify variations in UAV control signals to classify UAV type. To aid in these identification methods there is a growing data set of UAV control signals such as the DroneRFb-DIR, specifically developed to provide expanded data sources to conduct RF classification methods for UAV identification. While these identification classification methods are accurate and highly reliable in a controlled environment, RF-based identification systems are subject to failure due to external factors, including: interference, multipath fading or interference and congestion of the spectrum due to an urban environment. Generally speaking, these identification and classification methods require an extensive database of labeled data as well as extensive processor time to accomplish.

Deep learning using vision-based methods has become a significant method of using UAV's to detect UAVs. YOLOv8 is one popular choice due to its fast and accurate classification of UAV's and ability to be used in real time.

YOLO models complete UAV detection successfully versus using traditional detection methods, especially in dynamic situations. Using Deep Learning technologies have also been developed to differentiate between UAVs and birds, which has been a major barrier to successful visual detection. There are now advanced architectures, such as YOLO-FEDER FusionNet, that use enhanced feature extraction and feature fusion techniques to improve UAV detection capabilities. Visual-based systems are solely dependant on being in the correct environment. Lighting, weather and occlusion all change the perceived quality of the visual.

Blockchain-based authentication systems are a decentralized and tamper-proof way for UAV identity verification. Lightweight authentication protocols have been proposed to provide secure communication between UAVs and ground stations. Blockchain provides a secure way to share data and eliminates the requirement for a central authority, therefore reducing the possibility of being susceptible to cyberattacks. The use of blockchain to develop FANET

architecture provides a foundation of trust and security in the UAV network. However, the significant barriers to successfully deploying a real-time solution are computational overhead, latency and scalability.

Communication over the air in Unmanned Aircraft (UAV) networks has received much attention from researchers and various encryption methods have been proposed to secure the integrity and confidentiality of data transmitted between two or more UAV systems. For example, TinyJAMBU is an example of a lightweight cryptographic algorithm developed specifically for resource-constrained UAV systems [22]. Newer encryption mechanisms, such as homomorphic encryption and quantum-resistant (QR) cryptography, are able to provide stronger levels of security for UAV systems against newfound threats [20][21]. In addition, there have been successful improvements made to MAVLink protocol security utilizing encryption mechanisms in order to ensure secure communication between UAVs [19]. However, deploying effective methods of protection to UAVs that have limited resources is still a challenge.

Overall, both UAV detection and security mechanisms fall short of having an integrated format that combines detection, authentication, and response mechanisms together. This limitation further shows the need for a unified framework to encompass all of these features within a single system, like SkyShield.

2.10 HYBRID DETECTION SYSTEMS AND EMERGING TRENDS

In recent times, researchers have begun studying hybrid detection systems that use multiple detection methods to address the limitations of each detection method separately. For example, using hybrid systems that operate together with radio frequency (RF) detection, vision classification, and automatic radar detection enhances detection accuracy.

In addition, swarm intelligence and decentralized methods of coordination provide the foundation for collaborative detection and response using multiple unmanned aerial vehicle (UAV) systems. Energy-efficient and decentralized swarm coordination models increase the efficiency and scalability of systems [4]. Drone Swarm Decision Making (DSDM) algorithms facilitate the collective decision-making of UAVs or Drones in real-time, despite the fact that there may be a level of uncertainty in their environment [5].

Hybrid systems take advantage of the various detection methods available to produce a strong detection system. For example, while RF detection will identify the object, the vision detection will classify the object, and the radar detection will provide spatial awareness of the object and identify the object at long range. By combining these detection systems into a complete set of detections, hybrid systems provide a multi-layered approach that produces a more robust detection system with fewer false positives.

Artificial intelligence is a key component of hybrid systems because it allows deep learning models to be created that take data from multiple sensors and combine them into one dataset to allow for intelligent decisions to be made. By using edge computing platform technology, deep learning models can operate in real-time and rely on other than cloud technology to reduce latency and provide the drone with more autonomy.

Adopting blockchain technology will enhance hybrid solutions by creating a system where identity verification is secure and all transactions are recorded in an immutable ledger. Combining blockchain with UAV networks could potentially advance these systems' reliability and increase transparency, thereby establishing a decentralized way to manage trust in a given environment [23], [24].

While hybrid systems have certainly improved thanks to Blockchain, challenges still exist regarding the complexity of hybrid systems, sensor data synchronization, and computational load. The effective integration of different technology types remains an area of ongoing research.

2.11 LIMITATIONS IN REAL-WORLD DEPLOYMENT

While there have been numerous improvements in technologies to detect UAVs and ensure their security, there are also many obstacles to overcome before they can be used in the real world.

One of the most significant problems with these technologies is scalability. A lot of the existing technologies could not scale when there is a large number of commercial and/or military UAVs. As the number of UAVs increases in the airspace, it becomes difficult to establish communication between multiple UAVs based on current technologies. Network congestion and communication bottlenecks will likely not be resolved in the near term with the current software and hardware architectures used in most large-scale systems.

In addition to these challenges, there are also issues with unreliable and insecure communication among UAVs. UAV networks are extremely susceptible to cyber threats such as spoofing, replay attacks, and unauthorized intrusions. Although encryption techniques can improve the overall security of the network, the computational limits of UAVs often limit their ability to implement encryption protocols.

Additionally, environmental conditions affect the performance of UAV detection systems. Vision-based detection technologies are extremely sensitive to environmental conditions such as lighting conditions, weather, and the complexity of the background of a target. RF-based systems can also experience performance problems from the presence of signal interference or degradation.

While blockchain technologies offer a more secure method of communication between UAVs, the latency and computing overhead associated with the use of blockchain technologies may preclude them from being used in real-time applications. Furthermore, when using the public blockchain networks, transaction costs are associated with every transaction and scalability issues are common.

Finally, there are many ethical and regulatory issues associated with the use of counter-UAV technologies. Many counter-UAV systems rely on aggressive mitigation techniques such as RF jamming that disrupt legitimate communication systems and violate regulatory requirements.

Due to limitations of today's UAV defense systems, there is an increasing demand for an

integrated, scalable and efficient UAV defence system incorporating UAV detection, UAV authentication, and UAV response capabilities. To meet this need, the new SkyShield system combines edge AI with blockchain technology and multiple modes of sensing to produce a total UAV defence solution.

2.12 PROBLEM STATEMENT

There is a growing number of unmanned aerial vehicles (UAVs) being developed and used at an ever-increasing rate, which creates challenges for establishing a secure and controlled environment for unified airspace. While many detection and monitoring technologies have been developed and implemented in recent years, the majority of current systems operate independently rather than working together to establish safety and control over the airspace. Major subsystems that provide for the safety and control of the airspace (detection, authentication, and response) do not communicate with one another; therefore, these independent systems do not operate as efficiently as they could, have increased potential for threats to safety and security, and are not as easily scalable in terms of real-world applications.

In terms of current types of detection devices being used on UAVs, radio-frequency identification (RFID) is impacted by interference and congestion in urban environments, while visual classification devices are impacted by weather, light, and obstruction by other surrounding objects. Because these independent detection systems are impaired in their operation and functioning, both false negative and false positive occurrences will continue to exist and thus, detection systems will be unreliable.

Additionally, the vast majority of authentication systems used to verify the owner of a UAV are maintained in a centralized manner, which will allow them to be vulnerable to both spoofing types of attacks and manipulation of the data associated with the authentication as well as to a single-point failure (SPOF). Not having secure and widely created decentralized systems by which to authenticate the identity of individuals in situations where trust and accountability are critical will have significant negative impacts on the safety, security, and confidence of any individual or organization using a UAV. While blockchain technology presents a unique opportunity for securely managing data, the potential benefit of using blockchain technology for UAV systems is still poorly researched and understood.

Last but not least, one of the bigger issues with current counter-UAV systems is the reliance on physical countermeasures (e.g., RF jamming, physical destruction) to mitigate UAV-related security threats. Such techniques have a tendency to negatively impact legitimate users (e.g., pilots, air traffic controllers), as well as raise ethical and regulatory concerns. It is therefore critical to develop effective and controlled methods of addressing security threats while also ensuring safety.

In addition, the absence of unified graphical user interfaces able to integrate data from different sources (e.g., radar video, visual detection, authentication logs) limits the amount of information available for use by operators to manage their operations in an operationally effective manner. Lack of situational awareness also decreases the effectiveness of existing counter-UAV systems.

As such, the development of an integrated, comprehensive UAV defense system is essential in order to combine multiple modes of UAV detection, secure identity verification, real-time processing, and ethical response strategies into one cohesive framework. The SkyShield system is proposed to leverage edge computing, deep learning, blockchain technology, and multi-sensor integration to answer those needs through a scalable and highly efficient solution for airspace security.

2.12 SUMMARY

In this study, the literature review was carried out on recent and relevant literature in all aspects of multi-UAV collaboration, detection of rogue UAVs using different techniques, AI-based threat and payload analysis, secure communication, and blockchain for UAV verification. It has been observed that while individual aspects such as RF-based detection methods, swarm intelligence algorithms, and secure communication methods have achieved considerable progress, the overall literature in this area is still scattered and lacks cohesive real-time integration.

Most of the existing literature in this area has been focusing on individual aspects of detection, identification, mitigation, or authentication of rogue UAVs rather than addressing these issues as part of a unified framework for airspace security. This makes these solutions incapable of providing comprehensive solutions for end-to-end airspace surveillance, drone identification, ethical threat handling, and audit trails as part of a unified framework. In particular, while exploring the integration of blockchain and edge AI for resource-constrained UAVs, not much research exists in this area.

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Chapter 3

METHODOLOGY

3.1 INTRODUCTION

With the research gap highlighted in the literature review section, this research seeks to propose a novel system named SkyShield. The proposed system is intended to address the shortcomings of the existing fragmented Counter-UAV systems by integrating several cutting-edge technologies. For instance, the system utilizes swarm intelligence algorithms for effective coordination of drones. Secondly, the system utilizes radar technology for effective detection of aerial objects. Finally, the system utilizes blockchain technology for effective and unalterable verification of identities. Additionally, the system utilizes lightweight secure protocols for effective and efficient data exchange. The system also allows for seamless data exchange among the different technologies.

This concept is for a comprehensive, self-sufficient, ethical method for detecting and identifying hostile drone operations within no-fly zones. Additionally, this system's modular design provides for scalability so that the system can respond and adapt to various levels of threat and environments in which it is in operation. Each module can perform its required function as well as integrate efficiently with other modules due to a modular design philosophy. Such design concepts also provide the benefit of enhancing system maintenance and allow for future implementation of emerging technologies, such as artificial intelligence (AI) for threat categorization and responding; furthermore, the system has been designed and will conform to ethical and regulatory requirements.

3.2 SYSTEM ARCHITECTURE

There are 4 major components of the SkyShield architecture; each component interacts with the others to create a complete airspace defense system that is both autonomous and integrated. The main component is the Central Autonomous Drone, which is in charge of making decisions and carrying out mitigation actions against any unauthorized drones that enter the airspace. The Central Autonomous Drone is supported by a number of Support Drones, which have all been designed to include the ESP32-CAM, allowing live video stream capabilities and GPS and real-time tracking of any suspicious drone. Due to their collaborative operation, Support Drones create a distributed state that enhances the redundancy of the surveillance function.

A Ground Station is responsible for managing the drone system with a graphic user interface (GUI). The GUI is an application developed in Flask, which can be used by users to monitor the drones in a more effective way. The private Ethereum blockchain provides an authentication system to reduce trust issues related to financial transactions and to provide a

secure, transparent, and immutable identification of drones. The components of the drone system communicate through a very efficient communications protocol that uses both UDP and HTTP to provide high-speed performance for the system. Each communication channel within the system has used AES-128 for encryption to provide data integrity and data confidentiality to ensure that information transferred between the various components of the drone system will remain secure. The system was designed to be fault tolerant so that it can continue to operate even if one or more of its components fail. Consequently, the drone system will be much more robust and reliable when compared to other drone systems.

The overall system architecture represented by the block diagram, is based on a distributed and layered design in which all four modules (sensing, processing, verification, and response) will operate together. The Sensing Layer has both visual and spatial detection. Visual data will be obtained using camera modules (connected to Raspberry Pi and ESP32-CAM) and will be continuously monitored as long as the aircraft is in the airspace. The radar emulation module is constructed from an ultrasonic sensor and servo motor and periodically sweeps within an angular range to detect objects that may have come within the aircraft's detection range. This approach (dual-mode sensing) will increase detection accuracy through the use of two different types of data sources.

Data collected from the two sensing modules will be transmitted to the Edge Processing Unit (EPU) and processed in real-time. The primary processing will be conducted on a Raspberry Pi using the YOLOv8 deep learning framework to detect objects (UAVs) in each frame of data. Each misidentified frame will contain information regarding the identity of an object and its location via a bounding box; as well as the associated confidence level for detection. Processing at the Edge will reduce network latency and eliminate dependency on cloud processing by minimizing the amount of time needed for processing. By using EPU for processing, the overall system will be able to provide rapid detection and operate autonomously without cloud assistance.

The telemetry data of the UAV node will be transmitted to the ground station in real time with updates from the other components of the system. This data will include the UAV's identity, its positional data defined by GPS coordinates, and the time of the data point, and will be formatted as a JSON object for more efficient transmission of the data. To minimize the time it takes to transmit telemetry information, an extremely lightweight protocol such as UDP will be used for transmitting the telemetry, and an HTTP-based protocol (which provides a more reliable means for transmitting telemetry and interacting with the overall system) will be used for facilitating reliable transmission between the UAV nodes and the ground station. By using a hybrid communication approach, the system will maintain a balance between speed and reliability; both of which are particularly important for real-time drone monitoring applications.

Once the telemetry and detection data have been received, the system will proceed to the verification stage where validated UAV identities will be assigned using a blockchain. The Flask backend interacts with the Ethereum blockchain through Web3.py to make calls to smart

contracts that can be used to determine whether a drone has been registered. Each verification request creates a unique transaction that is stored permanently on the blockchain to provide transparency, prevent tampering with data, and eliminate dependence on centralized databases.

When the verification step is complete, the decision processing module combines the results of the detection and authentication processes and groups the detected drones into groups like authorized (legitimately), unauthorized (illegally flying), spoofed (atypically) based on established criteria, including both validation of identity and network behaviour. Based on this classification, the Centralized Autonomous Drone makes a determination as to how to proceed in executing a response action based on this classification made to mitigate threats in a controlled and effective manner.

The response action methodology of the system is based on the principles of being non-destructive and adhering to all applicable rules and regulations regarding safety (i.e., not employing aggressive counter-measures). Instead of using aggressive counter-measures as a method of slowing down or stopping high risk, suspicious, or prohibited activity, the system implements controlled responses such as the generation of alerts, tracking and making physical indications of the drone's location with a laser module. The controlled response approach provides effective management of the threat without the potential for additional damage or disruption due to unintended consequences from making an aggressive counter-measure.

All activity within the system is continuously monitored and visualized in a graphical user interface (GUI), which provides a central point of integration for data collected from many different subsystems. The GUI provides the operator with real-time information associated with the drone's location, radar detections, blockchain verification status and system alerts. The organization and display of the data in one visual format (dashboard) will increase the effectiveness of the situational awareness and will enable the operator to make an informed decision or take action with greater speed.

Furthermore, there is a design element included in the system architecture that will allow for providing feedback on how to improve the system as well as provides logging-of-events functionality that will document and maintain historical records for future retrieval and analysis for events or situations that are related to the detection results as well as the response that is taken by the system. All of the events associated with the detections of the drones and the associated responses to each event will be recorded on the blockchain so that there is a permanent record of each event that took place. This feature will assist in the event of needing to perform forensic and regulatory compliance.

The architecture of the SkyShield system is built on a tightly integrated and scalable framework that integrates sensing, intelligence, protection and response into one cohesive solution. The block diagram provides a visual illustration of how each unit in the SkyShield architecture interacts with one another and depicts the flow of the system operation in Fig. 3.1.

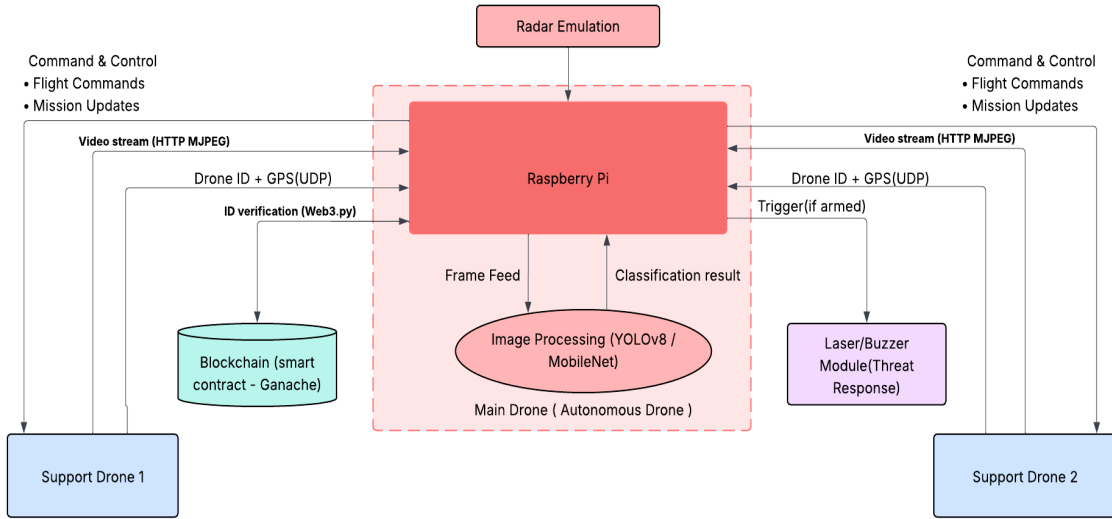


Fig. 3.1. Block diagram of SkyShield

3.2.1 HARDWARE COMPONENTS

The first prototype of the SkyShield is created from a collection of hardware parts that are carefully chosen for their low cost, widely available, and compatible with edge computing environments. The hardware design uses a distributed approach to architecture whereby sensing, processing, and communicating tasks are distributed across multiple devices in order to improve the efficiency of the system and the ability to scale. By using the distributed architecture approach, the SkyShield can operate in near real time by balancing the performance of the system with the utilization of system resources. Table 3.1 shows the complete list of hardware parts along with their assigned roles in the system.

The ESP32 microcontroller was chosen as the main node for both authorized and unauthorized drone simulation due to its built-in Wi-Fi feature, low power consumption, and adequate processing power for performing tasks such as building JSON packets and doing HTTP/UDP communications. The small size and included networking capabilities of the ESP32 microcontroller make it a good candidate for use in lightweight UAV nodes. In this case, two ESP32 DevKit v1 boards have been used—one board will represent an authorized drone transmitting valid identity and GPS data while the second board will simulate an unauthorized intruder transmitting spoofed identity data, thus allowing for realistic testing of the SkyShield's detection and verification capabilities.

The edge computing unit is the Raspberry Pi 5. It is capable of adequately performing as a YOLO v8 nano object detection model without the need for a dedicated graphics processing unit (GPU) and will process at approximately 14 frames per second (FPS). Utilizing Raspberry Pi enables real-time (RT) detection of objects at the edge and provides a low-cost and efficient option to achieve this.

The video streams from the IP Cam, when interfaced with a Raspberry Pi, are used to capture live video stream data, which enables the processing of videos into drone detection based

upon trained data models, providing continuous visual monitoring of airspace.

To mimic radar-based object detection, the ultrasonic sensor (HC-SR04) is used in combination with the SG90 servo motor to perform a basic two-dimensional (2D) angular sweep of the area being monitored for both measurement of distances to objects and their available spatial areas. A 5mW laser module interfaced to GPIO25 functions as a physical marker of a threat, activating when an unauthorized drone is present, and serve as a simple example of mitigation response mechanisms.

The entire SkyShield system is integrated within a ground station based on the use of a Windows laptop that runs the necessary software (including the Flask server and Ganache blockchain environment) and a real-time monitoring dashboard. The result is a cohesive hardware ecosystem that supports the end-to-end functionality of the SkyShield system, as described in Table 3.1.

Table 3.1 Hardware Components of SkyShield

Component	Model	Specification	Role in System
Authorized Drone Node	ESP32 DevKit v1	240MHz, 520KB RAM, WiFi 802.11 b/g/n	Transmits Drone_11 identity and GPS coordinates
Unauthorized Drone Node	ESP32 DevKit v1	240MHz, 520KB RAM, WiFi 802.11 b/g/n	Transmits ATTACKER_X identity to simulate intrusion
Edge Computing Unit	Raspberry Pi 5	Quad-core ARM Cortex-A72, 4GB RAM	Runs YOLOv8 inference at 14 FPS
Camera Module	Web Camera	8MP Sony IMX219, 1080p @ 30fps	Live video feed for YOLO drone detection
Laser Module	5mW Laser Diode	5V supply, GPIO25 signal pin	Physical threat indicator — ON when threat detected
Ultrasonic Sensor	HC-SR04	Range: 2–400 cm, 40Hz trigger rate	Radar distance measurement
Servo Motor	SG90	180° rotation, 4.8V operating voltage	Radar angular sweep for 2D scanning

3.2.2 SOFTWARE COMPONENTS

SkyShield's software stack makes use of open-source technologies that work together closely to provide effective inter-module communication (communication, visualisation, security and detection) within SkyShield's software architecture phases. These technologies have been chosen to provide seamless compatibility with edge devices while enabling scalability,

efficiency and cost-effectiveness in development. SkyShield has been structured on a modular design philosophy including the functionality assigned to individual components leading to improved clarity of systems, maintainability of systems and optimisation of systems performance. Table 3.2 provides a complete listing of SkyShield's software components along with their designated functions.

Python 3.10 is the primary programming language for creating the backend of the entire SkyShield system and for performing system orchestration because it has a wide variety of library support and can be easily integrated into a number of different types of operating systems. The Flask web framework is used as a lightweight web application that provides a place for users to monitor and visualise drone activity and interact with the data collected. Web3.py (the Web3 Python Library) creates the communication between the backend of SkyShield and the Ethereum network (the Blockchain used for performing Smart Contract transactions); Ganache is used by SkyShield developers to simulate a "sandboxed" local environment of the Ethereum Blockchain for testing and development; and Smart Contracts are deployed via the Remix IDE after being coded in Solidity. These components work together to ensure that the processes of verifying the identities of drones is secure and transparent.

Table 3.2 Software Components of SkyShield

Software	Version	Purpose
Python	3.10	Backend development and system coordination
Flask	2.x	Web server and DroneShield dashboard hosting
Web3.py	6.x	Blockchain communication via HTTP provider
Ganache	7.x	Local Ethereum blockchain simulator
Solidity	0.8.x	Smart contract development language
Remix IDE	Web-based	Smart contract compilation and deployment
YOLOv8 (Ultralytics)	8.x	Real-time drone detection model
PyTorch	2.x	Deep learning framework for YOLOv8
OpenCV	4.x	Video frame capture and processing
Leaflet.js	1.9.4	Interactive airspace map on dashboard
Chart.js	Latest	Threat distribution chart on dashboard
Arduino IDE	2.x	ESP32 programming and firmware uploading

TinyGPSPlus	Latest	GPS NMEA data parsing for future GPS module
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3.2.3 DATA FLOW AND INTER-MODULE COMMUNICATION ANALYSIS

The effectiveness of the SkyShield system is indirectly connected with the flow of the data between the various modules of the system. The architecture was designed for continuous and real-time data exchanges in order to allow all of the different components, or modules, to operate together in unison.

The flow of data commences at the sensor layer where raw data is collected through camera modules combined with radar sensors. This raw data is transferred to the edge processing unit where it goes through a two-step process of preprocessing and analytics. The video image frames are analyzed using the YOLOv8 model in order to detect and classify objects, and the radar data is used to determine the distance and direction of objects.

Telemetry information is transmitted to the ground station and is formed via drone identity, GPS locations, and timestamps and is structured to be processed in JSON format to allow for efficient parsing and processing of the data. Telemetry transmission uses UDP in order to provide lower latencies in the transmission whereas HTTP based communication is provided for reliable data transfer and the interaction with the system.

Upon receiving the telemetry and video data at the ground station, the backend system processes and integrates the input data from multiple sources. The integrated and processed data is forwarded to the blockchain module to verify the drones and ensure that only validated and authenticated drones may fly under the monitored airspace.

The backend system processes the incoming data from the ground station and combines data from several other sources. Once the data has been processed, it is sent to the blockchain module for identity verification of drones. This step ensures that drones who have been verified can access the area being monitored.

Once the drones have been verified, data is then sent to the decision-making module to analyze the data and develop a response. The results from this module are forwarded to both the response system and the visualization interface. The response system will take the appropriate actions and the visualization interface will refresh the dashboard with live data.

Through this continuous exchange of data between all modules, the entire system functions properly and can respond to potential threats in a timely manner. The use of lightweight communication protocols and structured data formats increases system performance by decreasing overhead and increasing processing time.

3.3 SYSTEM WORKFLOW

SkyShield's operational workflow functions as a systematic and sequentially organized methodology to provide for satisfactory disclosure, consolidation, and response to potentially

illegal UAV use cases. System wide (including subsystem components for sensing, communication, blockchain verification, and visualization), the SkyShield operational workflows establish a closed-loop architecture that permits real-time monitoring and control of the airspace.

The operational workflows begin with an initial initialization phase, which is comprised of all of the system components establishing connectivity, and preparing for operational use. This phase consists of both the UAV nodes (the ESP32 devices) connecting to a secure local WiFi network in order to communicate with the ground station, and the ground station establishing a private blockchain environment using Ganache, and deploying the smart contract that will be used for verifying the identity of the drone. A web-based graphical user interface that runs on the Flask application will also be initiated during this phase; this will serve as an integrated point-of-reference for monitoring every aspect of system performance. The initialization phase provides for the synchronization of all modules in the system, and verification that the communication links used within, and among, the modules have been established and functioning properly, prior to entering into the operational mode.

After the system has been initialized, it enters the real-time detection stage of the situational awareness cycle, which is where a radar emulation system will be continuously monitoring its environment using an ultrasonic sensor mounted to a servo motor. The servo will execute a two-dimensional angular sweep so that the system can detect any objects that exist in the area defined for detection purposes. The distance and angle to any detected object will be transmitted to the ground station and represented in the graphical interface as if they were radar signals.

In addition to the radar sensing data being sent to the Ground Station in real-time, telemetry data from UAV nodes will also be continuously transmitted to the Ground Station. This telemetry data will contain drone identification, GPS coordinates, and time stamps, and will be encoded using AES-128 encryption for security purposes. The telemetry data will be sent over low-latency communication paths using Lightweight Communication Protocols such as UDP and HTTP. The telemetry data will be used in conjunction with sensor data to create a complete model of the airspace being monitored by the Ground Station.

When data from detection and telemetry is ready to use, the subsequent step is for the system to verify the identity of an identified UAV. This process occurs when the ground station accesses the blockchain network in order to validate the UAV's identification with its UAV identification code number. Verification of the UAV's identification is made through the smart contract that had previously been deployed on the private Ethereum blockchain. The smart contract checks whether the drone identification code received by ground station has already been registered in the blockchain. If it is registered, then the UAV is considered "authorized" to operate in the airspace where it was detected, however if it is not registered, then it is considered "unauthorized" to operate in the airspace where it was detected. Each time that the UAV's identification is verified, a transaction is created in the Ethereum blockchain to create an immutable audit trail of every process conducted as part of this verification process.

While simultaneously verifying the identity of the UAV through its UAV identification code number, the ground station also initiates the video surveillance module to provide the operator with visual confirmation of the identified UAV. The video from the ESP32-CAM or Pi camera modules is sent to the Flask-based interface using HTTP-based MJPEG streaming so that the operator can quickly view the identified UAV and visually confirm that the UAV is accurately detected. When visual confirmation is provided to the operator with respect to whether the identified UAV has been detected, it provides an additional layer of verification for both the automated detection and verification processes. The combination of the information from the sensor data and from the visual confirmation of the UAV will help to improve the overall accuracy of the UAV detection and reduce the number of false positives.

In the decision-making and response phases, the first action is to identify the Authorized or Unauthorized Drone Classification, according to Drone Classification, which will determine the required response. The system performs passive monitoring and logs activity for Authorized Drones, and does not take any action. When there is an Unauthorized Drone present, the response actions determined by system logic will cause the establishment of: activation of the Central Autonomous Drone (CAD); generation of alerts; and initiation of Tracking protocols. Each of these response actions will be recorded on the blockchain in order to provide transparency and accountability.

The last component of the Workflow will be to visualize and log data. The GUI serves to consolidate data from all subsystems and display this information in a consolidated, easy-to-read format, via a dashboard. The dashboard will include subsets for Radar Scans; GPS Positions of Drones; Verification; Live Video; and System Event Logs. The Blockchain will include additional analytical data relating to Airspace Violations; Rogue UAV identification; and timestamps of Verification Events. The result of the complete visualization of Data is that the Operators can maintain an awareness of what is happening and make decisions based on real-time data.

The workflow provided by the SkyShield System represents an efficient, well-structured, fully-integrated system that provides sensing; communicating; verifying; and responding under one operating system, seamlessly in Fig. 3.2. Each of these steps is executed in a sequential manner to provide dependability in terms of the detection of UAV threats, secure verification of user identification, and efficient mitigation of UAV threats—which are all needed for implementing the system in security-critical environments that will require it to be operationally deployed.

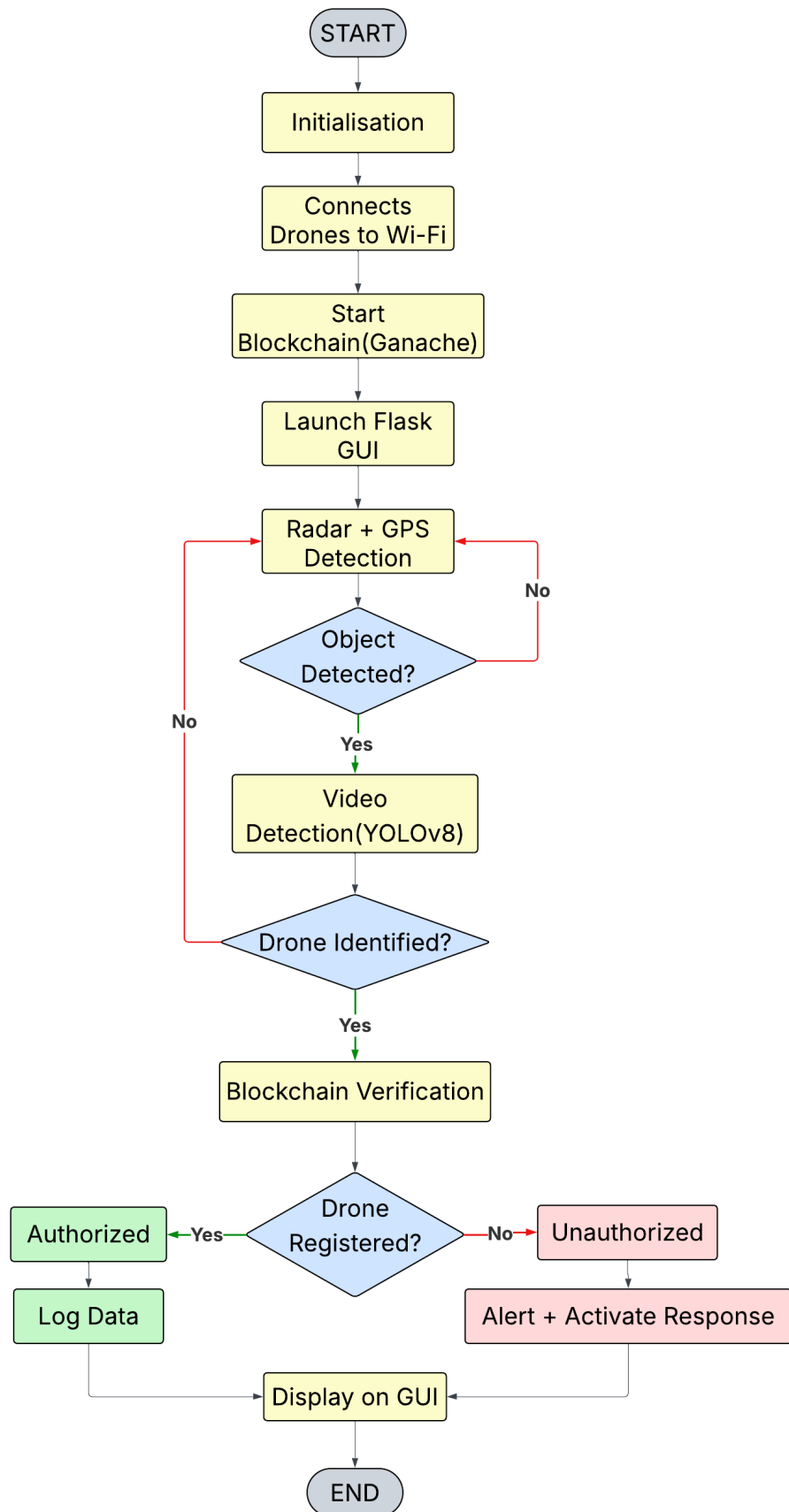


Fig. 3.2. System Workflow Flowchart

3.4 COMMUNICATION PROTOCOLS

The GPS position and identity of a drone at any given moment (in the form of packets) are transmitted as telemetry data over a wireless link between the ESP32 nodes and ground station using the User Datagram Protocol (UDP). UDP allows for low-latency communication, as it is a non-connected (or connectionless) protocol. By using a non-connected protocol like UDP, the transmission overhead can be kept to a minimum and telemetry data delivered more quickly. This is crucial for providing real-time responsiveness during changing airspace conditions. The responsive characteristic of UDP will assist in providing timely tracking and decision-making requirements in all UAV-based monitoring systems. In addition, the fast characteristics of UDP may serve as a loss (packet acknowledgement) tolerant protocol, so the packet would be delivered very quickly, but with some packets possibly being lost, since UDP does not provide a guarantee of reliable delivery of packets. Therefore, the trade-off between a performance-based system where responsiveness is prioritized over the importance of packet delivery reliability is achieved, as shown in Table 3.3.

The video stream from the ESP32-CAM module is sent to the graphical user interface using standard HTTP based MJPEG streaming protocol. This way, the video frames can be sent as consecutive JPEG images using standard web protocols (HTTP) and allow easy integration with the Flask maintained GUI. The way MJPEG is used to stream video has less computational overhead than other video streaming methods, making it ideal for embedded systems like ESP32-CAM modules. As a result, the system provides operators with timely (smooth), near real time visual feedback, which is an important part of assessing and monitoring threats accurately. The communication mechanism, along with its function in the overall system, is detailed in Table 3.3.

The Skyshield Identity Verification system uses Web3.py to connect to the private Ethereum Blockchain (over HTTP-protocols) in order to facilitate efficient interactions between the back-end and the smart contracts used for verifying if a drone is truly authentic. All of these authentication methods allow for secure and tamper-resistant evidence for authentication. The identification of UART communication serves as a means to exchange data directly and reliably between hardware components—primarily between the radar module and ground station. The use of Flask-based APIs allows for structured data exchanges, command executions, and control functions to occur between the GUI and the central autonomous drone. In summary, these various communication methods ensure that data transfer between the elements of the system occurs in a timely, capacity-balanced, and secure manner; thus, the exact roles each of these communication methods serve related to their interactions with the devices in question will be outlined in Table 3.3.

Table 3.3 Communication protocols and device interactions in the proposed system

Purpose	Protocol	Devices
GPS/ID Transmission	UDP	ESP32 → Ground Station
Video Streaming	HTTP MJPEG	ESP32-CAM → GUI
Blockchain Verification	Web3.py + HTTP	GUI ↔ Blockchain
Radar Data Transfer	UART Serial	Radar → GUI
Command Dispatch	Flask API	GUI ↔ Central Drone

3.5 SYSTEM IMPLEMENTATION OVERVIEW

The proposed SkyShield system is a completely integrated UAV defense solution. The proposed solution includes the YOLOv8 algorithm for drone detection, Ethereum blockchain for identity verification and radar emulation using an ultrasonic sensor and servomotor. The Haversine formula will be used to provide GPS based localization, and the real time web dashboard will provide the user with a real time web interface. The completed proposed solution is designed for full autonomy but still provides for human in the loop monitoring via the real time web dashboard interface.

Two ESP32 microcontrollers have been working to simulate an authorized and unauthorized drone and each microcontroller transmits telemetry to ground control over a secured Wi-Fi connection. The purpose of this experiment is to show how authorized and unauthorized drones could be tested within realistic environments. The authorized drone is set with a laser module attached to GPIO25; when the drone detects an unauthorized drone, it can activate this laser module, therefore providing a very basic example of physical countermeasures to unauthorized drones being detected by authorized drones. This integration of active threat response capabilities into the authorized drone system demonstrates functionality above and beyond simply monitoring activity. All verification processes performed on the drones have been logged on the Ganache blockchain using a unique hash for each transaction to create an immutable, uniformly accountable record of the data regardless of how it is modified. The combining of these technologies creates a valid approach to developing and delivering an affordable, efficient solution to countering UAV threats.

3.6 SYSTEM IMPLEMENTATION

3.6.1 CLASSIFICATION MODEL – YOLOV8

YOLOv8 is used for performing object classification and localization within the UAV detection module. YOLOv8 provides an excellent balance between accuracy and processing speed, making the model suitable for real-time detect-tracking of drone applications. The YOLOv8 architecture has been specifically designed for edge computing environments where computational power is limited but requires very low latency with high levels of responsiveness. Therefore, YOLOv8 is well-suited for deploying in the SkyShield System, primarily on platforms like Raspberry Pi. The efficacy of the trained model is further

demonstrated through the summary of performance metrics found in Table 3.5.

In addition to its high level of accuracy, several advanced features within YOLOv8 also make it an even better choice for real-time applications. One of these features—anchor-free detection—approaches detection differently from previous versions of YOLO, as it does not use predefined anchor boxes to help predict center points for detected objects; therefore, adapting YOLOv8 to different object sizes and shapes enhances the model's overall ability to generalize and produce accurate results. Furthermore, YOLOv8 enables multiple computer vision tasks (i.e., object detection, instance segmentation, and image classification) to be accomplished by the same model and framework. Consequently, the extensive variety of tasks performed supports future improvements of the SkyShield System without needing to make significant architectural alterations.

Table 3.4 YOLOv8 Custom Dataset Summary

Parameter	Value
Total Images Collected	847
Training Set	677 images (80%)
Validation Set	170 images (20%)
Number of Classes	1 (drone)
Annotation Format	YOLO format (.txt bounding boxes)
Input Image Resolution	640 x 640 pixels
Annotation Tool	LabelImg (open-source)
Camera Used	Raspberry Pi Camera Module v2
Collection Environment	Indoor and outdoor settings
Augmentation Applied	Horizontal flip, brightness variation, rotation

Table 3.5 Performance Metrics

Metrics	Score
Accuracy	0.87
Precision	0.89
Recall	0.87
F1-Score	0.88

One of the most significant benefits of utilizing YOLOv8 is that it is both scalable and lightweight in terms of user-defined configurations. In other words, there are several different versions of the YOLOv8 architecture available (from Nano to Extra Large), which allows developers the freedom to choose the configuration that best fits their available computing power. The current implementation selected a Lightweight Model variant to ensure maximum efficiency on Edge devices without compromising the overall performance level of the object detection model being used for UAV object detection. With a user-friendly development interface (Command Line Interface (CLI) and Python Application Programming Interface (API)), developers can easily perform all phases (Training, Validation, Deployment) of their object detection system's development lifecycle quickly and efficiently.

To develop a custom UAV object detection model, we built a custom dataset of drone imagery captured from different view angles and weather conditions. Using bounding boxes in the YOLO format to accurately represent the location of drones within each photographed image; we annotated each of the drone photographs in our custom dataset. We then divided our annotation dataset into two subsets, Training and Validation, so that we could evaluate our model's generalization ability to the validated data and avoid overfitting. Table 3.4 provides a detailed description of the training dataset's characteristics (number of images, resolution of images, and augmentation methods), which demonstrates the strength and diversity of the dataset utilized to train our model.

A configuration file called `data.yaml` was created to facilitate the training process which contained paths to datasets; a list of class labels; and additional parameters associated with each class. All of the dependencies required for the training (Ultralytics YOLO, PyTorch and NumPy) were managed via a requirements file to provide the required level of reproducibility. The model training process was carried out using appropriate hyperparameters, such as the number of epochs; batch size; and input resolution, to determine how the model was trained to fit the specific training data. During the training process, the model's checkpoints were saved to retain the model's weights with the best performance as determined by validation performance. The trends in training and validation performance are illustrated in Figure 3.1, which shows the rate of convergence and stability of the model.

After the training and validation had been performed, the optimised YOLOv8 model was deployed in real-time scenarios for inference on both static images and live video feeds, simulating a real-time common UAV detection scenario. Inference was performed using custom inference scripts that processed incoming frames and produced outputs (bounding boxes) along with the corresponding confidence scores. This phase of the project validated the model's ability to perform well in real world settings and was able to show that the model is capable of being used in nearly real-time operation.

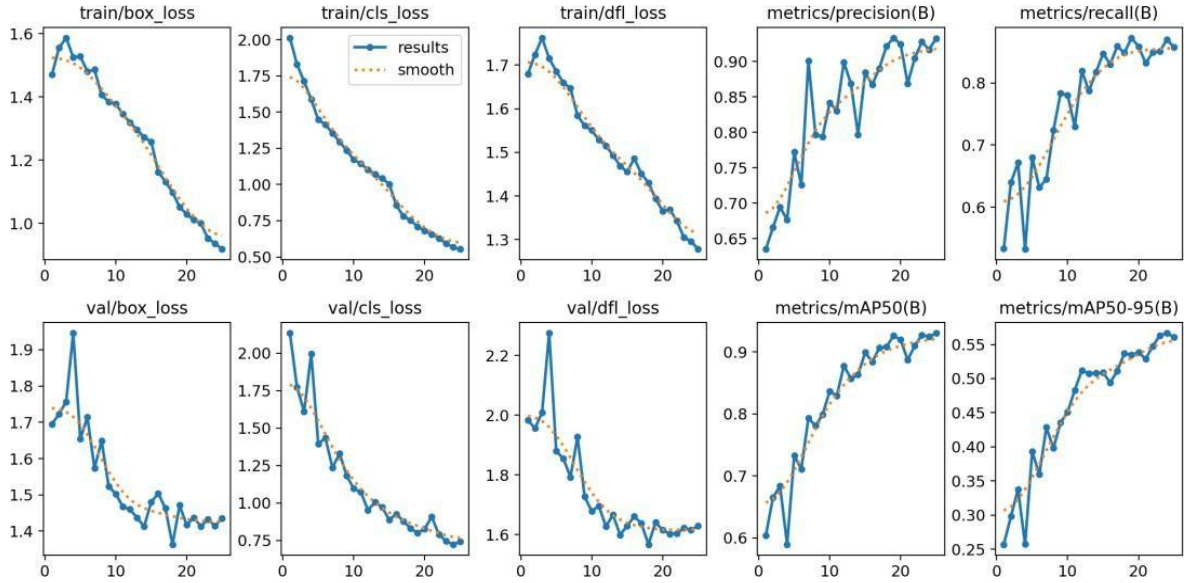


Fig. 3.3. YOLOv8 Model Training and Validation Metrics



Fig. 3.4. Detection results of the trained YOLO model showing identified drones

The inferences from the results demonstrate that UAVs can be accurately detected as well as given appropriate confidence levels indicating that the detection module is an effective detection method for UAVs. An example of the detected drones (by being highlighted) can be seen in Fig. 3.4. Optimization was performed on multiple levels during development on how the system would operate most efficiently, this included eliminating issues with pathing/annotation; mismatched configurations; and other issues pertaining to the dataset. In addition, optimizations were also performed to increase the speed of inference while maintaining a high detection rate to allow smooth operation on edge devices without

degrading performance. The information shown in Table 3.4, Table 3.5, Fig. 3.3 and Fig. 3.4 shows that the YOLOv8-based detection module within the SkyShield system has reliability and is an effective detection solution for UAVs.

3.6.1.1 DATASET PREPARATION

A dataset of custom drone detection was created from a range of photos captured from various perspectives, distances and lighting conditions to ensure the dataset is robust for use in real-world applications. In total, 847 photos of different types of drone activities, including in-flight and stationary, were collected at varying sizes relative to the camera's position making the data diverse enough to allow for learning of spatial and contextual variance of patterns in this dataset. Photos were taken with the Raspberry Pi Camera Module v2 in both indoor and outdoor environments, making the dataset representative of how a model may perform in the real world by capturing images in multiple operational conditions.

By including variations like background clutter, illumination changes and partially occluding objects to improve the quality of the dataset, the annotation process was completed using Labellmg, a popular open-source image annotation program. Every instance of a drone was labeled using rectangular bounding boxes in YOLO format which encodes all relevant information necessary for a model's training including a label for each object type along with normalized x and y coordinates for the centre of an object and its width and height. This enables compatibility with the YOLOv8 training pipeline. To achieve a proper evaluation of the model and avoid overfitting, the dataset was split into training (80% of photos) and validation (20% of photos) subsets consisting of 677 and 170 images respectively. As previously stated, the details of the dataset will be found in Table 3.4.

3.6.1.2 TRAINING CONFIGURATION

The YOLOv8 nano (yolov8n.pt) model was used for this UAV detection project because it is a lightweight model that can be used on an edge device like the Raspberry Pi 4. The model was initialized with weights from the COCO dataset and then was fine-tuned on our custom drone dataset through transfer learning, which decreases training time and increases the model's accuracy. This is possible because the model can use the features it has learned on a very large dataset.

The training process employed an optimal hyperparameter set for optimal performance in terms of both accuracy and computational efficiency. The hyperparameters were adjusted for epoch no, batch size, input image, and learning rate corresponding to dataset and hardware constraints. A Stochastic Gradient Descent (SGD) optimizer with momentum was used to provide stable convergence during training. The loss functions used during training for bounding box regression was Complete IoU and for classification was Binary Cross-Entropy.

All model training was performed on a 4GB Raspberry Pi 4 and did not utilize any dedicated GPU, showing that edge-based model training is feasible. Training time was approximately 2.5 hours; during each training session, weights of the models were saved periodically based on validation loss to keep track of the best performing weights. A summary of all

hyperparameters and configurations used for the experiment can be found in Table 3.6. This table provides an overview of the entire training process and optimization strategy.

Table 3.6 YOLOv8 Training Hyperparameters

Hyperparameter	Value
Model Architecture	YOLOv8n (nano) — lightweight for edge deployment
Number of Epochs	50
Batch Size	16
Input Image Size	640 x 640 pixels
Initial Learning Rate	0.01
Optimizer	SGD with momentum 0.937
Pre-trained Weights	COCO dataset (yolov8n.pt) — transfer learning
Hardware Used	Raspberry Pi 5, 8GB RAM (no GPU)
Approximate Training Time	~2.5 hours
Best Model Checkpoint	Saved at lowest validation loss epoch
Loss Function	CIoU for bounding box, BCE for classification

3.6.1.3 JUSTIFICATION FOR SELECTING YOLOV8 OVER OTHER DETECTION MODELS

When developing a UAV detection system that operates in real time, it is important to choose the right object detection model. In this case, YOLOv8 has been selected over other deep learning models due to its ability to perform better in terms of both accuracy and speed and also being a better fit for edge deployment (i.e., running on low-resource devices).

While traditional object detection models such as R-CNN, Fast R-CNN, and Faster R-CNN have had a reputation for being able to accurately detect objects, objects are detected with these models using a two-stage detection process where regions are proposed first before they're classified as detected objects. The two-stage pipeline makes it less efficient from a computational perspective and increases the latency when handling real-time applications; therefore, it is not practical to implement the R-CNN, Fast R-CNN, or Faster R-CNN model for real-time applications using low-resource devices like Raspberry Pi.

Single Shot Detectors, such as the earlier versions of YOLO including YOLOv3 and YOLOv5, provide fast detection speeds but are limited with respect to how well they can detect small objects or maintain accuracy when detecting objects against complicated backgrounds. These are significant performance constraints in UAV detection scenarios, especially considering that drones, which are usually very small relative to their distance from the UAV detection system can result in not being able to detect the drones completely.

YOLOv8 represents a major step forward in the development of YOLO networks. The new network architecture includes several important improvements that improve object detection for small or distant UAVs and are able to drive higher levels of precision and recall than previous YOLO models while allowing for multi-task learning of object detection/classification/segmentation in a single unified framework.

The newest feature of YOLOv8 is a completely new anchor-free detection method that eliminates the dependency on anchor boxes by allowing the YOLOv8 object detector to adapt to the size and shape of an object being detected, thereby resulting in a much higher level of detection accuracy.

In addition to the anchor-free detection capability of the YOLOv8 model, other improvements include a new optimized backbone for feature extraction, improved techniques for feature fusion, and improved process for capturing spatial and contextual information. Collectively, these improvements combine to produce the highest level of performance ever achieved by YOLO models, providing an optimal balance between speed and accuracy.

The small version of YOLOv8, called YOLOv8n, is specifically targeted to perform real time processing on edge-based devices without the use of a GPU in order to provide a suitable solution for implementing embedded systems such as the Raspberry Pi employed in this project.

The choose YOLOv8 for the development is an additional important aspect; is a user-friendly to implement and integrate via the Ultralytics framework. The framework provides an intuitive API for the training, validation and deployment of YOLOv8 reducing the overall complexity of development. Additionally, pre-trained weight and transfer learning make it significantly easier to develop with YOLOv8.

YOLOv8 is also much faster and requires fewer system resources compared to transformer-based models like DETR (Detection Transformer). This makes YOLOv8 more applicable for real-time applications as well. On the other hand, transformer models can perform very well in some cases, however their high resource demand limits their application in edge based systems.

Ultimately, YOLOv8 was selected as the best model for the SkyShield system based on its high accuracy in detecting objects, capable of performing in near real-time, adaptable for small object detection, and can work in edge computing environment. Because of these characteristics, YOLOv8 is the best model to use for detecting UAVs in a dynamic resource-limited environment.

3.6.2 BLOCKCHAIN AUTHENTICATION

The blockchain authentication module is tasked with securely identifying and authenticating UAVs (unmanned aerial vehicles) using an Ethereum-based smart contract deployed on a local Ganache network. The primary goal of this particular module is to authenticate all detected drone identities against a tamper-proof and decentralized ledger, thus removing the

threat of having a centralized and modifiable database. The system also utilizes blockchain technology to guarantee the integrity, transparency, and traceability of all events associated with the drone authentication.

In order to support the secure and real-time authentication and monitoring of UAVs, the proposed system contains both software and hardware components. For example, Python 3 will be used for back-end processing and system management; Flask will provide a web-based (i.e., graphical user interface) for monitoring the UAV in real time, as well as enabling human interaction with the UAV. Ganache is configured to provide a local environment for developing and testing smart contracts on a local Ethereum blockchain. Additionally, Web3.py is used for providing a seamless communication interface between the Flask back end and the Ethereum blockchain. Finally, the Arduino IDE will be used to program and configure ESP32 microcontrollers that will provide reliable telemetry communications from the UAV back to the ground. Collectively, these software components form the overall UAV authentication framework described above in Table 3.2.

Two ESP32 devices are used as part of the hardware for the purpose of representing an actual/fake drone (gift) using a real-life example of a UAV. One ESP32 is a real drone (i.e., authorized), and the other ESP32 is an unauthorized drone (i.e., An intruder). These two ESP32 devices are sending their telemetry data across an HTTP network over a shared Wi-Fi network to the ground base. Telemetry data from both drones includes the Drone ID, GPS coordinates, and a timestamp. The real drone has a laser connected to GPIO25. The laser will be activated when the drone perceives a threat, and the act of activating the laser may indicate to the person on the ground that a response has occurred. The ground base is a laptop that has both the Flask server and the Ganache blockchain host. The hardware components fit into the system architecture illustrated above in Table 3.1 to support the seamless interconnection between the physical and digital systems.

To initialize the system, the DroneRegistry contract must be deployed on Ganache using Remix IDE. The Ganache local blockchain environment used for deployment is illustrated in Fig. 3.13. After, the authorized drone must be registered using the registerDrone() function. Next, the Flask backend must connect to the blockchain via Web3.py. Finally, both of the ESP32 devices must be configured with the server's IP address so they can continue sending telemetry data to the backend while active. Once the system is operating, all telemetry is transmitted via real-time monitoring, which can be viewed on the dashboard Fig. 3.5, Fig. 3.6 showing live blockchain activity and verification logs respectively.

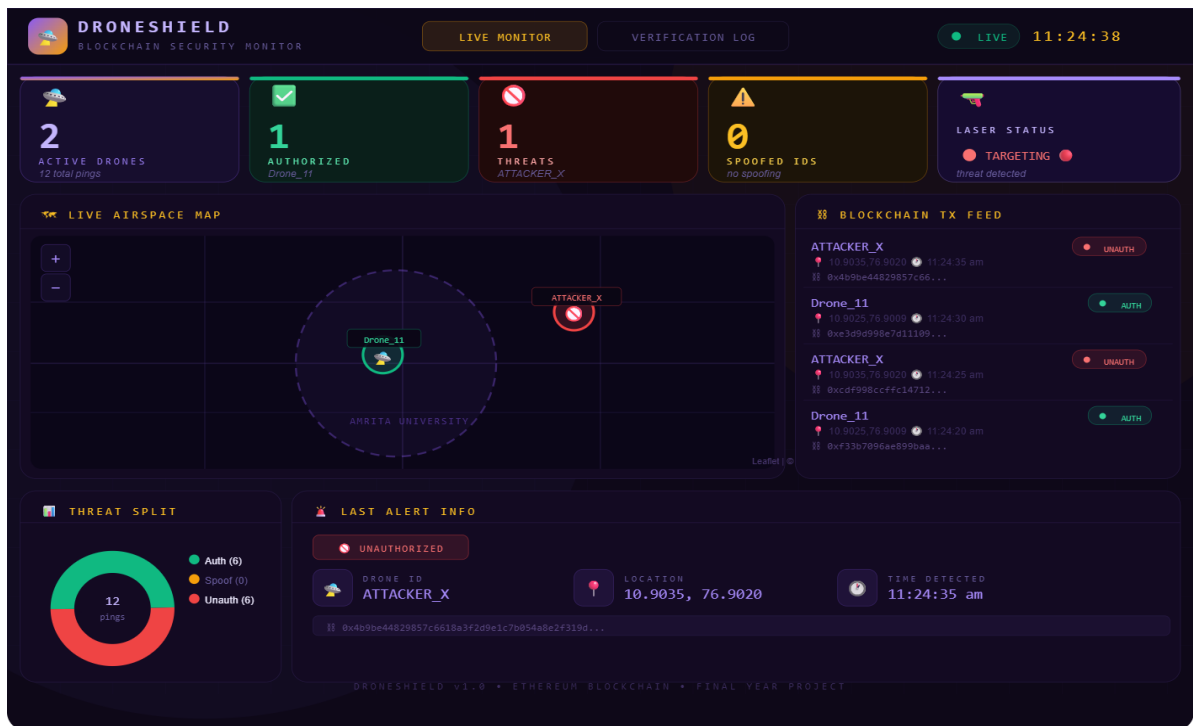


Fig. 3.5. Live Monitor for Blockchain

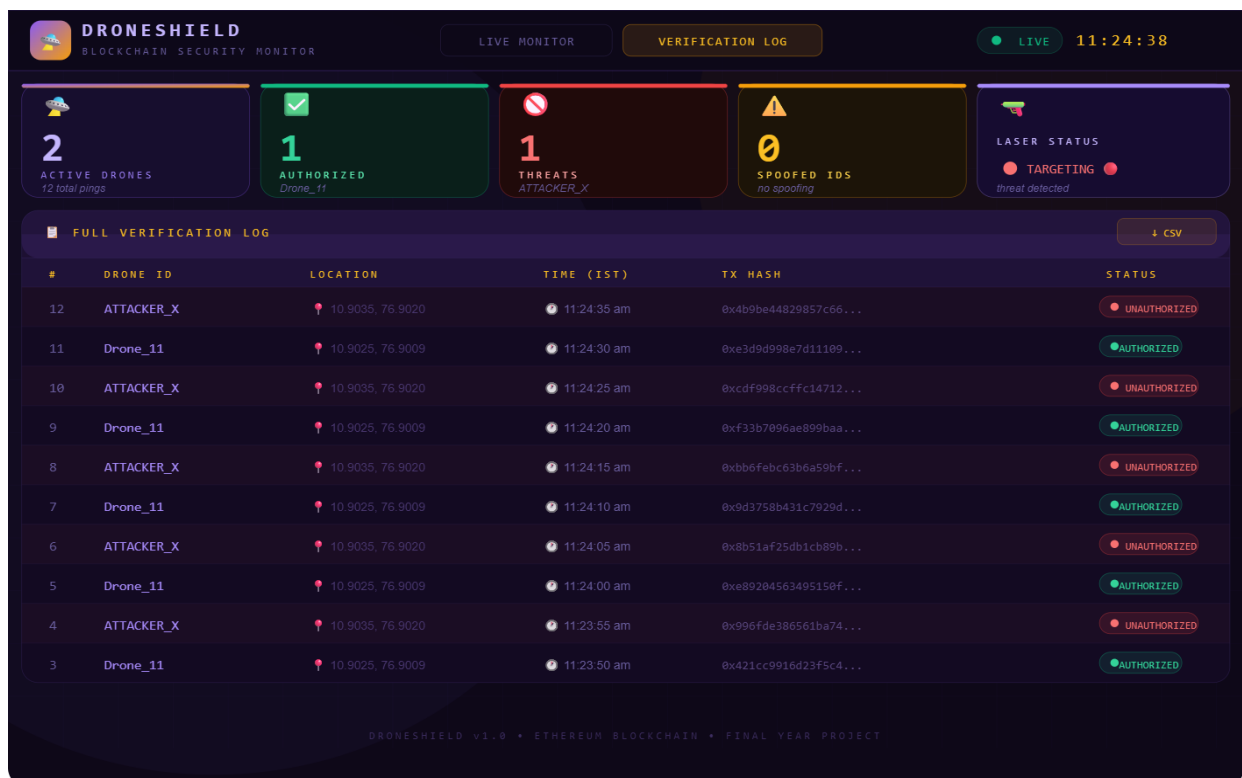


Fig. 3.6. Verification Log

Continuous and secure data exchange between ESP32 nodes, Flask back-end, and the blockchain network is how the operational process is done. ESP32 periods will send structured JSON packets with drone ID, GPS coordinates, and a Unix time stamp through HTTP POST requests. The Flask back-end will then execute the `verifyAndLog()` function of the smart contract deployed on the blockchain to verify the identity of the drone, and all verification attempts will be stored in an immutably fashion on the blockchain.

Along with the processes already mentioned, spoof detection works via the association of an IP (internet protocol) address to a drone ID. If a drone ID is received from any different IP than that of the drone's ID, the drone will be classified as having been impersonated, i.e., an identity has been cloned. After applying this rule to all drone IDs, the drones can be classified into three different categories, Authorized Spoofing, Spoofed Identity, and Unauthorized Drone. This classification provides one more layer of protection than the current method of simply checking to see if a drone is registered.

Each time you verify an event, there will be an Ethereum Transaction Hash issued as a unique ID for the event. This hash is your digital certificate as it holds verifiable information of the event and will remain on the Ethereum blockchain indefinitely. The Flask dashboard will display key information such as the drone's current location and authorization status (which authorizes the drone to operate), a transaction log showing each transaction that occurred on the blockchain, and alerts related to any threats detected.

When the Detection Module identifies an unauthorized (or spoofed) drone as having entered into the vicinity of the authorized drone, the System is programmed to activate the Laser Module (also known as the Thrust Reversal) on the authorized drone. This allows for a physical response to an unauthorized event. The combination of detection, verification, and response constitutes a closed-loop security system that is fully functional.

The Blockchain Authentication Module functions in tandem with the YOLOv8 detection pipeline and the overall architecture of SkyShield. Once the drone is identified visually through the camera feed, its identity will also be verified on the blockchain in real-time. Both visual detection and the identification through blockchain will appear simultaneously in real-time on the Graphical User Interface (GUI). This system enables the subsystem to operate in real-time, enhancing the reliability, responsiveness and practical application of the SkyShield System on a global level.

To set up the DroneRegistry system, you must first deploy the smart contract onto the Ganache network using the Remix IDE. After deploying the smart contract, the next step is to register authorized drones onto the blockchain by calling the `registerDrone()` function. Once the drones have been registered on the blockchain, the Flask API server this will be connected to the Ganache network using Web3.py and the ESP32 modules will be configured to point to the IP address of the Flask server for sending data. After the configuration of all these system components is complete, the dashboard will be started so that drone activities can be monitored, visualized, and logged in real time.

The system is designed to have a continuous, secure exchange of data between the ESP32 modules, the Flask-based backend server, and the Ganache blockchain network. The system begins with each ESP32 module transmitting structured JSON packets with drone ID, latitude, longitude and current unix time to the Flask backend servers every five seconds using HTTP POST requests. Once the Flask backend receives the data from the ESP32 module, it calls the `verifyAndLog()` function of the `DroneRegistry` smart contract to verify the drone's registered status on the blockchain.

The system is able to identify identity spoofing by recording and storing the first IP address that sends the ESP32 device with a specific drone ID. If another ESP32 device sends the same drone ID but has a different IP address than the original device, this indicates that the drone is attempting to spoof or clone the identity of a legitimate drone. The system then categorizes threats according to how the verification was done: registered drone ID sent from the original device → Authorized; registered drone ID sent from a different device → Spoofed; unregistered drone ID → Unauthorized.

For each verification, a unique hash (Ethereum transaction hash) will be generated, providing proof of the verification that is recorded for all time on the blockchain. The Flask web dashboard will provide continuous updates to the live map of airspace, the authorization status of the drone in that airspace, the feed of blockchain transactions made regarding that drone, and any thread alerts associated with that drone. Upon the identification of an unauthorized or spoofed drone, the authorized drone will activate its laser module by way of GPIO25, providing physical evidence of the target drone to the user.

The integration of the authentication on the blockchain with the `DroneShield` web dashboard, detection via YOLOv8, and hardware through ESP32 is complete. The acoustical detection via YOLOv8 with camera confirmation of the drone visually allows the ESP32 to send the drone's identity to the Flask backend for verification on the blockchain simultaneously. Once verified, the result is reflected on the live airspace map, the transaction feed on the blockchain, and last alert panel on the dashboard and consists of three possible states (Authorized, Spoofed, or Unauthorized). The integration of visual detection, identity verification, and laser engagement comprises the full operational loop of the `SkyShield` system.

3.6.2.1 SMART CONTRACT DETAILS

In the Solidity 0.8.x programming language, the `DroneRegistry` smart contract has been created and run on the Ganache blockchain through the Remix IDE. The functional interface of the deployed smart contract is shown in Fig. 3.14. The primary function of the contract is to use 2 main data types: a `drones` mapping (which contains information such as drone id #, owner name, registration timestamp, active status) and a `flightLogs` dynamic array (which records all attempts to verify drone identity). These two structures provide an efficient method to look-up drone by their unique id, and to create a permanent record of all verification attempts using the blockchain.

The `registerDrone()` function is an administrator-only function and uses the `onlyAdmin` modifier to restrict drone registration to these users only. This function takes in drone id # and owner name parameters and saves these values to the mapping, along with the current block timestamp. The `verifyAndLog()` function can be called publicly and will verify any drone in the registry and log the verification attempt to the `flightLogs` array (regardless of the verification result) in order to have a permanent record of all interactions with each drone (including any attempts to create unauthorized drone access).

In Tables 3.7 and 3.8, a summary of the functions and their roles has been provided, along with an orderly representation of how the smart contracts work as well as the performance metrics of the blockchain when they were implemented (gas, transaction confirmation time, and network configuration).

Table 3.7 Smart Contract Functions

Function	Access Level	Parameters	Description
registerDrone()	Admin only (onlyAdmin modifier)	droneId (string), ownerName (string)	Registers an authorized drone in the blockchain registry with block timestamp
verifyAndLog()	Public — called by Flask backend	droneId (string), location (string), timestamp (uint256)	Checks drone registration and permanently logs the verification event on-chain
drones()	Public — read only	droneId (string)	Returns full drone registration details: owner name, active status, registration time
flightLogs()	Public — read only	index (uint256)	Returns a specific flight log entry: drone ID, location, timestamp, status

Every time you create an Ethereum transaction hash with `verifyAndLog()`, you can find the hash by going to the Transactions tab of the Ganache interface. The gas consumed for each transaction was 65,000-80,000. The average confirmation time was 120-150 milliseconds. Therefore, the blockchain-based authentication system has been shown to be efficient and suitable to use for real-time UAV actions.

Table 3.8 Blockchain Transaction Performance on Ganache

Metric	Observed Value
Average Gas Used per Transaction	65,000 – 80,000 gas units
Average Transaction Confirmation Time	120 – 150 milliseconds
Blockchain Network	Local Ethereum — Ganache v7.x
Smart Contract Language	Solidity 0.8.x
Deployment Tool	Remix IDE (web-based)
Transaction Visibility	Ganache GUI — Transactions Tab
Log Storage Type	Permanent — flightLogs[] array (immutable on-chain)
Admin Account	w3.eth.accounts[0] (Ganache Account 0)

Each call to `verifyAndLog()` generates a unique Ethereum transaction hash. In the Ganache environment, the average gas used per transaction was observed to be approximately 65,000–80,000 gas units, with an average confirmation time of 120–150 milliseconds. All transactions are visible in the Ganache GUI under the Transactions tab, providing a transparent audit trail.

3.6.3 RADAR EMULATION WITH GUI

The Radar Emulation module delivers an interactive graphical output of UAV activity and integrates sensor and verification data, captured by the ESP32, to create a visual representation of UAV activity via an interactive User Interface that demonstrates how a conventional radar sensor would function. The module will operate like a radar sensor, using a servo motor for angular motion and an ultrasonic sensor to measure distance. The servo motor will provide the continuous rotational sweeping motion, while the ultrasonic sensor will measure distances to other objects that may be classified as aerial targets within the total distance of the servo motor sweep. Figures 3.5 and 3.6 will show the physical configuration of the emulation module and how the sweeping mechanism operates.

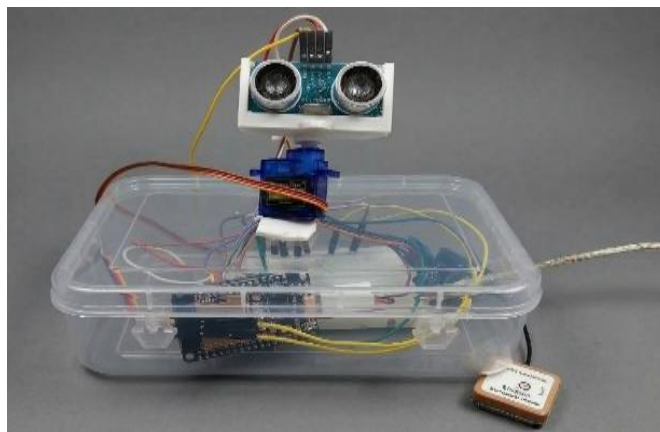


Fig. 3.7. Ultrasonic Sensor with Servo Motor

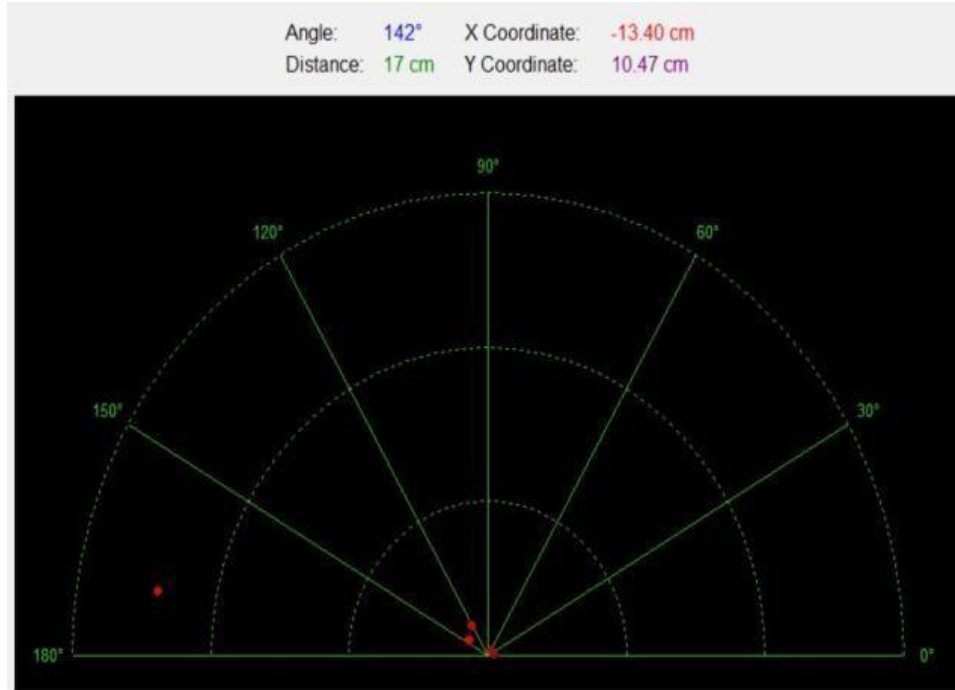


Fig. 3.8. Radar Sweep

The data collected from detecting the angle and distance associated with each potential detected elevation, once transmitted to the backend system, can then be processed (e.g., in Python) and presented in a radar-style by using a graphical user interface created with Tkinter's GUI framework. Each object detected will be presented as a dynamic blip on the radar interface, and the entire radar display will be continuously updated based on the movement of the servo motor. In addition, the overall visual output of the system is enhanced through the use of the blockchain verification result data integrated with the radar visualization itself. The visual output will display classified (by color) regulated aircraft (i.e., authorized) and non-regulated (i.e., unauthorized or suspicious) aircraft to allow for immediate operator alerting in the case of non-regulated aircraft. GUI provides extra information in addition to radar representations like detection angle, measured distance, and verification time stamps providing a complete picture of the current environment. Radar data and real-time detection output and system log data are integrated, as illustrated in Figs 3.7 and 3.8, which show both the SkyShield console and the real-time event logs. Combining both spatial sensing and identity intelligence to improve operator situational awareness through a single display.

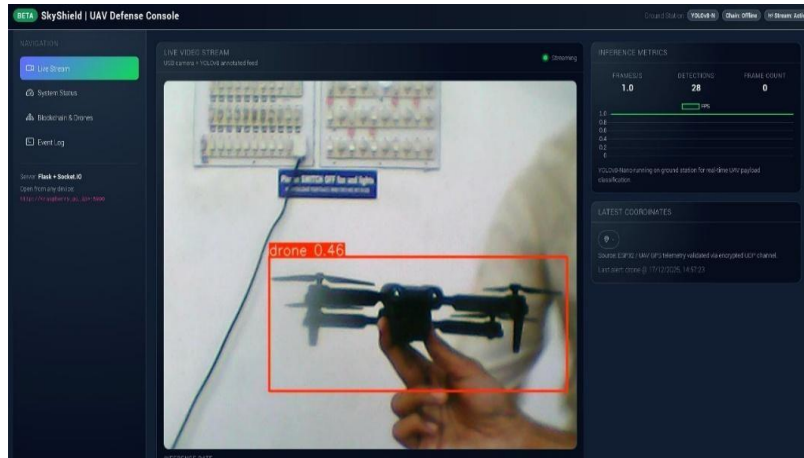


Fig. 3.9. SkyShield UAV Defense Console with Live YOLOv8 Inference



Fig. 3.10. Real-Time System Event Log

3.6.4 RADAR TO GPS LOCALISATION AND DISPATCH MECHANISM

After the radar emulation module finds and detects the object, the system will begin the localization process, which converts distance-based radar measurements into a usable form of geographic coordinates. The ultrasonic distance sensor provides measurement of distance only; as such, the angular position of the servomotor provides an angular position relative to the ground station and the combination of these polar coordinates gives the approximate Cartesian coordinates. Subsequently, by using the known location of the ground station, the approximate Cartesian coordinates can be converted into the GPS coordinates.

The GPS coordinates that have been obtained through the process above will be sent to the Raspberry Pi via low-volume and UDP type of data communications. The Raspberry Pi, acting as an edge-processing device, will check the location information that has been received and then send the location information to the central drone control station. Based on the location

information received from the Raspberry Pi, the drone operator can now fly the drone towards the located object to perform further inspection, follow-up, or neutralize the object. This process provides the necessary link between initial detection and action.

The Haversine validation mechanism is used to provide a way to improve the accuracy of the localization process, which allows the GPS coordinates to be estimated after radar data has been collected. The Haversine formula is used to measure the distance of detection by radar between the ground station and the UAV in a geodesic manner accounting for the curvature of the Earth and to provide a more accurate distance than would be able to be measured using a planar approach alone. An implementation for this validation process can be found in Fig. 3.11. This step will provide more reliability in the navigation process, assistance with the zonal classification of threats, and also assist to ensure that the trajectory tracking of an individual UAV is consistent prior to taking action in response to a threat.

```
---- HAVERSINE METHOD GPS ----  
Home Location   : 10.877, 76.898  
Target Location : 10.89, 76.9088  
Distance        : 1.86556 km
```

Fig. 3.11. Haversine-based distance calculation

3.6.5 BLOCKCHAIN-BASED UAV DETECTION DASHBOARD

By using a blockchain-based UAV detection dashboard, users can quickly see the airspace being monitored in real-time, including visual data from YOLOv8 detection (radar) and GPS tracking (i.e., display of location). The dashboard provides specific information regarding each detected UAV—GPS location, authority to fly in the airspace being monitored, and classified threat assessment of the detected UAV as authorized, spoofed, and/or unauthorized.

Smart contracts are going to be crucial in classifying threats posed by UAVs that have been detected – for example, an unregistered UAV enters a monitored or controlled area; a match between the drone’s ID and its IP address has been made (identity spoofing); and replay attacks caused by reusing the same IP address or any unauthorized UAV entering a controlled area after having had its right to fly suspended. The Drone Registry Smart Contract will provide a source to validate the identity of each detected UAV, ensuring that no UAVs that are unauthorized or have had their right to fly revoked will be allowed to continue to access the airspace. All UAV threat events and other detected UAVs will be stored permanently on the blockchain to provide verification sources for future reference.

The dashboard interface shown in Fig 3.10 will allow real-time visualization of all airspace activity and associated threat alerts, and the status of the verification of those threats. An overview of the underlying blockchain environment and the related transaction logs to the dashboard can be seen in Fig 3.11, and an overview of the smart contract interface that will be incorporated into the dashboard can be obtained from Fig 3.12. Each event documented on the blockchain will include a set of detailed metadata fields: the UAV's ID, IP address, type of threat, block number, and date and time of the event, allowing for a complete forensic analysis of each event documented on the blockchain. The complete SkyShield UAV Airspace Security Dashboard is shown in Fig. 3.12.

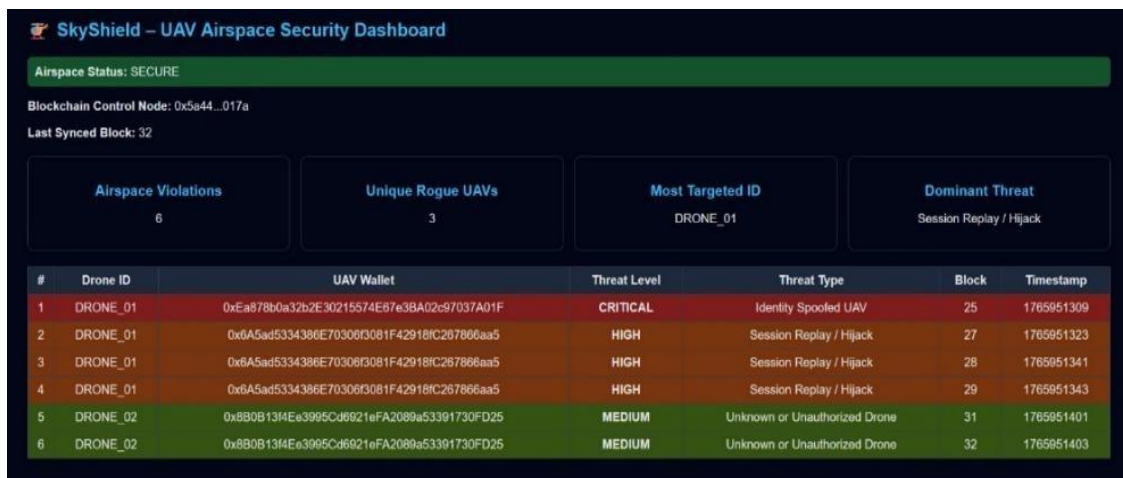


Fig. 3.12. SkyShield UAV Airspace Security Dashboard

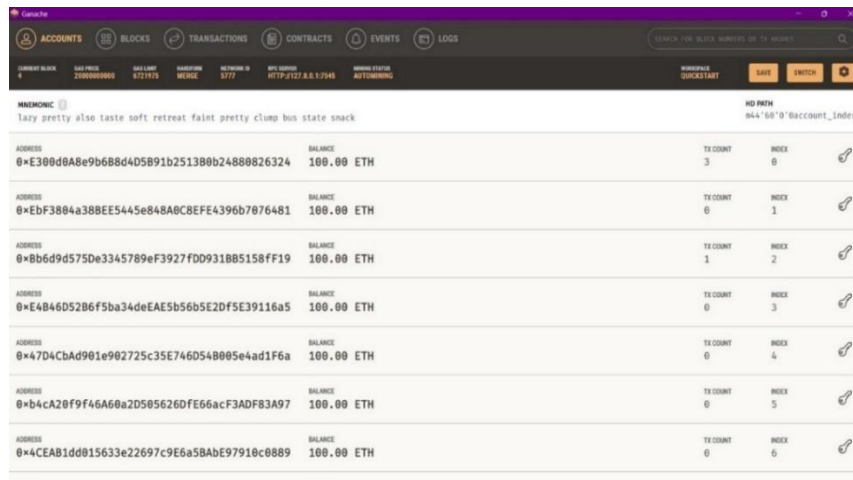


Fig. 3.13. Ganache Local Blockchain Environment

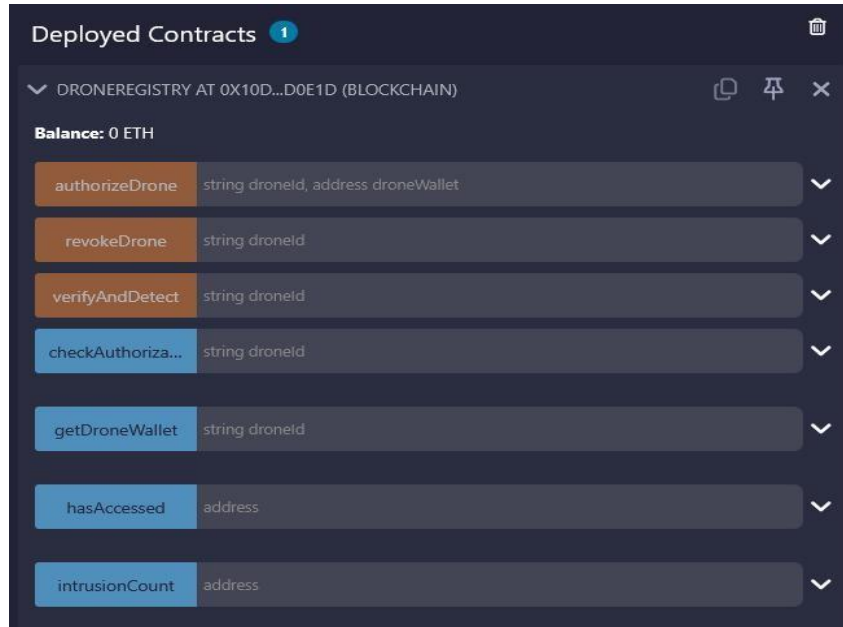


Fig. 3.14. Smart Contract Functional Interface

In addition, the hardware configuration of the edge inference technology associated with this system is provided in Table 3.9 which describes the functions of the Raspberry Pi, camera, YOLOv8 detection model, and blockchain backend. This highly integrated architecture allows the detection, verification, visualization, and response components of SkyShield to operate together to create an effective and scalable UAV defense system.

Table 3.9 Edge inference hardware configuration

COMPONENTS	DESCRIPTION
Computing platform	Raspberry Pi running Linux-based edge node for local video analytics.
Camera interface	Raspberry Pi Camera Module providing real-time video stream to the detection pipeline.
Detection model	Custom YOLOv8 drone detector converted to OpenVINO IR format for NCS2 deployment.
Blockchain backend	Private Ethereum blockchain (Ganache) with Drone Registry smart contract for ID verification.

3.7 FULL SYSTEM INTEGRATION

The SkyShield System is comprised of tightly integrated and coupled modules that work together in real-time as an end-to-end pipeline to create a comprehensive solution for UAV defense. The Architectural design captures the entire cycle of information processing: sensing, detecting, verifying, localizing, responding creating one integrated workflow with the lowest latency and highest operational efficiency. Each of the individual systems has distinct functionality; however, they all share continuous exchange of information with each other

creating a closed loop autonomous defense system.

Radar emulation modules will perform 2D sweeps using Ultrasonic Sensors and Servo Motors interfaced with ESP32 continuously identifying aerial objects around them and transmitting distance and angle relative to ground stations. This provides initial situational awareness for potential targets and allows confirmation of the target through the use of the YOLOv8 Model that analyzes live video data from the ESP32-CAM or Pi Camera Module usually providing confidence values between 0.7 and 0.92 via visual identification of drones in real-time. The combination of radar sensing and artificial intelligence-based vision has resulted in much improved detection and reduced false positive rates as shown above in fig 3.7.

The ESP32 is used to send structured JSON packets containing drone ID, GPS coordinates and Unix Timestamp to the Flask API over HTTP POST requests at regular intervals after drone detection. The Flask API will call the `verifyAndLog()` function on the deployed DroneRegistry smart contract in Ganache to verify the identity of the received drone. Drones are categorized as Authorized, Spoofed, or Unauthorized based on the verification logic specified in the blockchain authentication subsystem (see Table 3.7). This verification process will help support the risk-based threat classification strategies outlined in Table 3.10, providing multiple methodologies for mitigating threats from various methods of attack.

The derived polar coordinates from the radar will be converted into GPS coordinates and validated by using the Haversine formula to determine the geodesic distance between the two sets of coordinates. Therefore, if the drone location is validated accurately, then its localization can be determined with assurance (refer to Figure 3.9). The validated drone location will also be used for tracking, threat assessment, or potentially sending a centralized autonomous drone to the location of the identified drone to take further action.

All captured data is displayed on the DroneShield Flask dashboard, which provides the main visual interface in real-time. The dashboard serves as the connecting point for multiple sources of data, such as live airspace maps, blockchain transaction histories (including their unique transaction hashes), threat statistics and alerts. Examples of the visualizations produced by the system are illustrated in Figures 3.10 and 3.11, respectively. If the system detects unauthorized/spoofed drone activity, it will trigger a physical response by activating the permitted drone's laser module via GPIO25 to demonstrate a real-world mitigation technique.

Verification and response records are stored directly in the blockchain, creating a permanent, immutable record that can be used for auditing purposes. All security threat response capabilities (including unauthorized access detection, identity spoofing detection, replay attack and data integrity attack) are identified in Table 3.10. The combined abilities of the sensors, intelligence, verification and response components create the SkyShield as a robust, scalable unmanned aircraft vehicle (UAV) defense system.

Table 3.10 Security Threat Analysis

Threat Type	Description	SkyShield Detection Method	System Response
Unauthorized Access	Unknown drone ID not registered in blockchain registry	Smart contract checks drones mapping — returns unregistered status	Classified as Unauthorized — laser activated, red alert on dashboard
Identity Spoofing	Legitimate drone ID transmitted from a different attacker ESP32 device	Flask compares ESP32 IP address against first registered device IP	Classified as Spoofed — laser activated, orange alert on dashboard
Replay Attack	Old valid packet retransmitted by attacker to gain access	Unix timestamp validation — stale timestamps flagged	Logged with warning — operator alerted via GUI event log
Tampered Records	Attempt to modify or delete flight log entries after recording	Blockchain immutability — records cannot be altered once confirmed on-chain	All logs permanently stored — tamper detection automatically ensured

3.8 MATHEMATICAL MODELING OF SKYSHIELD SYSTEM

Several mathematical modeling methodologies have been integrated into the SkyShield system to facilitate accurate detection, localization, and decision making.

3.8.1 HAVERSINE DISTANCE CALCULATION

The Haversine formula can calculate the geodesic distance between two points located on earth's surface. It is given by:

$$d = 2R * \arcsin(\sqrt{(\sin^2((\varphi_2 - \varphi_1)/2) + \cos\varphi_1 * \cos\varphi_2 * \sin^2((\lambda_2 - \lambda_1)/2))})$$

Where:

- R is the radius of the Earth
- φ represents latitude
- λ represents longitude

This formula uses the earth's curvature in the calculations, and will generate an accurate estimation of distance for GPS-based localization.

3.8.2 RADAR COORDINATE TRANSFORMATION

The radar module provides distance (r) and angle (θ). These polar coordinates are converted into Cartesian coordinates using:

$$x = r * \cos\theta$$

$$y = r * \sin\theta$$

These coordinates are then mapped to GPS coordinates using reference offsets from the ground station.

3.8.3 YOLOV8 LOSS FUNCTIONS

The YOLOv8 model uses:

- **CIoU Loss** for bounding box regression
- **Binary Cross Entropy (BCE)** for classification

The Central Intersection over Union (CIoU) is a metric that improves the accuracy of localization by considering the overlapping area of the target objects as well as the distance between their centers, and the ratio of their aspect ratios.

3.9 ALGORITHM DESIGN

The SkyShield system has an established systematic mathematical model to create a systematic mathematical model for determining UAV identification, verification, and mitigation ("Identify the drone") in real-time. The decision-making function consists of three individual, yet cooperating together, core algorithms for establishing: (1) drone identification, (2) detection of spoofing, and (3) threat classification. The three algorithms operate in a sequential and cooperative manner to provide for accurate drone detection and secure drone classification in the monitored airspace.

The blockchain technology serves as the basis for the drone verification algorithm that will verify a specific UAV's validity. Secondly, the Spoof Detection algorithm increases security by analysing an IP address or source of an identity clone attempt to prevent your identity from being cloned via spoofing methods on any network. Lastly, the Threat Classifications algorithms will use the outputs of the detection and verification modules to identify what the best response action would be. A full description of how each of the three algorithms operates is now provided below.

3.9.1 ALGORITHM 1: DRONE VERIFICATION

This algorithm provides secure authentication for UAV Identity through communication with the blockchain based UAV Registry. All UAVs detected by the system must be validated against pre-registered credentials to determine whether or not they have been authorized to be located there.

Input: UAV ID, GPS Coordinates, Timestamp

Output: UAV Authorization Status (Authorized or Not Authorized) and Transaction Hash

Procedure:

1. The ESP32 node receives telemetry data from the UAV, including the UAV ID, GPS coordinates, and timestamp.
2. This telemetry data (UAV ID, GPS coordinates, and timestamp) are formed into JSON

format for processing.

3. The ESP32 node sends a request to the blockchain network through the use of the Web3.py interface by invoking the verifyAndLog() smart contract function.
4. The smart contract then finds the UAV ID within the UAV Registry on the blockchain network.
5. If the UAV ID can be found, the UAV is considered "Authorized"; otherwise, the UAV is considered "Unauthorized."
6. A unique transaction hash will be generated for the authentication request.
7. The outcome of the authentication request including timestamp and GPS coordinates will be stored permanently on the blockchain network.
8. The outcome of the UAV authentication request and the transaction hash will be returned back to the back-end system and displayed on the dashboard.
- 9.

This algorithm provides authentication that cannot be tampered with and eliminates the need for a central system to manage user identities.

3.9.2 ALGORITHM 2: SPOOF DETECTION

The Spoofing Detection Algorithm detects identity spoofing attacks by checking whether there are any inconsistencies with the drone ID and the IP address of that drone.

Input: Drone ID, Current IP Address

Output: Spoofing Status (Normal or Spoofed)

Process:

1. The system receives telemetry packets with the drone ID and current IP Address.
2. The system checks for a local mapping of that drone ID in its database.
3. If the drone ID has never been mapped, then create a new entry in the mapping database with the current IP address as a valid address and set it to a status of normal.
4. If the drone ID has been mapped before, look up the prior (valid) address in the mapping database.
5. Compare the current IP address with the prior IP address in the mapping database.
6. If the two IP addresses match, the drone ID will be marked as "Normal."
7. If the two IP addresses do not match, the drone ID will be marked as "Spoofed," indicating a possible attempt to replicate that ID.
8. When a drone ID has been marked as a Spoofed ID, an alert will be sent to the operator's dashboard, and the incident will be recorded with timestamp and IP information. This process provides additional protection from attacks designed specifically to bypass traditional verification processes.

3.9.3 ALGORITHM 3: THREAT CLASSIFICATION AND RESPONSE

The Threat Classification algorithm integrates outputs from detection, verification, and spoof detection modules to determine the threat level of each UAV and execute appropriate response actions.

Input:

- YOLO Detection Output (confidence score, bounding box)
- Verification Status (Authorized / Unauthorized)
- Spoof Status (Normal / Spoofed)
- GPS Coordinates

Output:

- Threat Level (Authorized / Spoofed / Unauthorized)
- System Response

Procedure:

1. The YOLOv8 model provides detection and confidence score.
2. If the confidence score is lower than a pre-defined value, the detection is ignored as false Positive
3. The Drone Verification algorithm provides verification status.
4. The UAV will be categorized based on specific standards that include:
 - If the UAV passes the verification process and no false identity has been established, it is classified as Authorized;
 - If the UAV passes the verification process but false identity has been established, it is classified as Spoofed;
 - If the UAV does not pass the verification process, it is classified as Unauthorized.
5. Once classified, the appropriate response to the UAV will occur as follows:
 - Authorized - No action will occur and only the event details will be saved for record and history.
 - Spoofed - A warning alert will be generated for monitoring purposes and the laser module will be activated as a controlled response.
 - Unauthorized - A critical alert will be generated, the laser module will be activated and tracking will begin.
6. The actions taken and the results of all classifications will be recorded onto a blockchain system for access to audit & traceability verification for all authorized users.
7. The dashboard is also updated with the drone's location, threat level and system issued response in real time.

This algorithm enables intelligent decision-making by combining multiple data sources and ensures a structured and ethical response to UAV threats.

3.10 DATA FLOW AND SEQUENCE ANALYSIS

SkyShield architecture has a standard process of data flow:

1. ESP32 sensors gather telemetry data.
2. Telemetry data is sent from ESP32 sensors to the Flask back_end server.
3. The Flask back_end server receives telemetry data and interacts with the blockchain to determine if the telemetry data has fraudulent activity (i.e., data with a smart contract).
4. Flask back_end server sends response data back to GUI.
5. GUI displays the results of the fraud detection process.

SkyShield architecture uses asynchronous communications to enable the system to operate nearly real time for continuously monitoring the detection of fraud and thus minimizing latency in the detection of fraud Events.

3.11 SYSTEM SECURITY ARCHITECTURE

The security architecture of SkyShield integrates multiple layers:

- AES-128 Encryption ensures confidentiality
- Blockchain ensures integrity and immutability
- IP Mapping prevents spoofing
- Timestamp Validation prevents replay attacks

This layered approach provides robust protection against cyber threats.

3.12 SUMMARY

SkyShield UAV defense system offers an integrated and total capability to address the most pressing security challenges of the airspace environment through the application of Artificial Intelligence (AI), blockchain technology, and "Internet of Things" (IoT) based sensors. The SkyShield system is a single operational pipeline for processing all the subsystems associated with UAV detection and mitigation, ensuring timely, accurate, and reliable aerial threat detection, verification, monitoring, and response.

The SkyShield system's superior threat classification and detection capability provides the basis for threat identification and mitigation from multiple attacks/disruptions as detailed in Table 3.10. SkyShield can detect/mitigate unauthorized access, identity spoofing, replay attacks, and data tampering. The identity spoofing detection provides additional security to the system in that it associates the drone ID with a unique IP address at the drone's initial registration. Any discrepancy with this mapping can immediately be flagged as an event, providing an additional layer of authentication beyond just blockchain verification and enhancing security in a dynamic and changing network environment.

Using blockchain technology allows for verification of drones, including the verification process and flight logs to be safely stored and verifiably immutable. Unlike traditional centralized storage, a decentralized ledger ensures the integrity, transparency, and accountability of the recorded events. The ability to provide assurance during a forensic

investigation, to ensure all transactions are permanently recorded, with a unique cryptographic hash number assigned to each transaction, provide regulatory compliance for the aerospace industry as well. The effectiveness of real time monitoring and event tracking through the GUI interface (an illustration is shown in Figure 3.8) provides evidence of the level of monitoring done on a drone's operations.

SkyShield consists of one autonomous main drone, several support drones, and a blockchain-enabled ground station that are connected by a secure communications network from an overall system architecture perspective. SkyShield's communication network utilizes UDP to provide low latency telemetry data; MJPEG over HTTP for real-time video streaming; and Web3.py over HTTP for blockchain interactions. A complete overview of how each communication protocol is used can be found in Table 3.3. All three protocols provide for the effective, reliable, and scalable exchange of data between all components of the entire system.

The five primary components of the implemented system intersect to provide a complete functioning defense for unmanned aerial vehicles. The YOLOv8-based detection uses real-time visual detection and produces a verified detection performance of approximately 87% based on the information provided in Table 3.5. The module responsible for providing secure identity verification through blockchain provides verification of identity with average latency measured at approximately 120-150 milliseconds per transaction, as presented in Table 3.8. The radar emulation module employs ultrasonic sensing to provide two-dimensional awareness of space, and the Haversine-based localization module accurately determines geodesic distances and assigns the detected threat to the correct classification to monitor and track the threat.

In addition, the DroneShield Centralized Web Dashboard integrates all system outputs into a single common point of reference, including live visualization of airspace, blockchain transaction logs, and alerts that detect threats, as provided in Fig. 3.12. Supporting edge hardware configuration that enables the complete real-time inference and coordination of the system is summarized in Table 3.9 and includes the Raspberry Pi, camera module, and detection pipeline.

All in all, SkyShield shows that it is possible to create a highly scalable, energy efficient, and secure airspace defence system for unmanned aerial vehicles (UAVs) that will allow continuous monitoring, intelligent threat detection and prevention, and reactive threat mitigation with respect to threats from UASs. The technology used within the SkyShield solution, such as AI-based detection, blockchain-based authentication, and IoT-enabled sensing, provides a sound basis to build on as we continue to develop autonomous Airspace Security Systems.

Chapter 4

RESULTS AND DISCUSSIONS

4.1 YOLOV8 DETECTION PERFORMANCE

To evaluate and train a model based upon YOLOv8, a new UAV-specific data set had to be constructed for this research, which allowed for testing to finish producing results, as shown in Table 3.5. Through the analysis performed, the YOLOv8 model had an overall accuracy of 87%, precision of 89%, recall of 87%, and an F1 score of 88%. The Loss Curves for both training and validation converge at approximately 20 epochs, showing that Learning was stable with minimal amounts of overfitting. The gradual increase in mAP@0.5 throughout training also indicates the model developed new capabilities over time, allowing for better identification of UAV objects.

When used in Real-time applications, the YOLOv8 model successfully identified UAVs in video captured from live feed cameras, with the confidence associated with the detections ranging from 0.70 – 0.92. Overall Detection performance has been consistent in normal light conditions, yielding stable detection outcomes throughout various sample test cases. Detection performance dropped slightly in cases of very low light levels and with partially hidden features of the drones where they could not be completely seen. Even with these limitations, the YOLOv8 model maintained a low false-positive rate in instances of background features exhibiting a structural similarity to detected UAVs. Overall, these findings support continued use and research into use of the YOLOv8 model as a viable solution for real-time UAV detection along the edge, provided there is sufficient lighting and proper placement of cameras on the detection target(s).

4.2 BLOCKCHAIN AUTHENTICATION PERFORMANCE

In order to assess accuracy, latency & strength; the test process involved multiple drone identity transmissions. A total of 15 drone identity packets were verified for use in authorized, spoof or unauthorized attempts. All 15 packets were correctly identified as either Auth, Spoof or Unauth, yielding a 100% verification accuracy rate. Thus, the logic of the used smart contract in the DroneRegistry module is confirmed as functioning properly - as shown in Table 3.7.

Latency while verifying packets on the blockchain was measured locally (Ganache) to approximately 120 - 150 milliseconds per transaction, which corresponds directly to the results shown in Table 3.8. Each verified packet generated its own unique hash, providing traceability/auditability of all verification actions. Additionally, all intrusions attempts (Spoofing & Unauthorized Access) were captured & recorded immutably with an associated date/time stamp, drone ID, IP address & Threat Type classification. Therefore, not only will

the blockchain module properly secure the validation of identity, but it will also provide a reliable forensic logging mechanism for the operation of UAVs.

4.3 RADAR EMULATION AND LOCALIZATION PERFORMANCE

Using an ultrasonic sensor and servo motor, the radar emulation module was tested for several criteria: detection range, angular resolution, and real-time responsiveness. In successful tests, the radar-emulation unit was able to detect objects at distances exceeding 200 cm; therefore, it is suitable for short-range indoor surveillance applications. The system's servo motor can provide adequate angular resolution to estimate the positions of detected objects (approximately 3°).

The integration of radar-derived polar coordinates, GPS-based positional information (i.e., location on Earth), and Haversine distance validation methodology enabled a significantly improved accuracy in the estimation of the position. distance measurement will vary slightly based on the reflective property of ultrasonic waves and other factors affecting the distance reflectance reference point.

A Tkinter-based graphical user interface has been created to allow for real-time visualisation of detection angles, distances, and authorisation status. enabling persons who view the radar data from the user interface to easily understand the information being reported on as detected. Based on the results of this research, a very low cost radar emulation system used in conjunction with computational validation methodologies has demonstrated that it can provide the user with a practical solution for providing accurate infinitesimal spatial awareness, see Fig 3.9.

4.4 SYSTEM LATENCY AND REAL-TIME PERFORMANCE

The latency of the SkyShield system's real-time performance was quantified at each stage of the processing pipeline. The radar detection module has approximately a 50ms latency, while the YOLOv8 inference stage on the Raspberry Pi has an average latency of 120ms. The blockchain-based identity verification has an additional latency of approximately 150ms, and the latency caused by GUI updates is approximately 40ms.

The total combined latency of the system from end to end is between 400ms and 500ms, as shown in Table 4.3. The latency levels are acceptable for short-range unmanned aerial vehicles detection and monitoring applications, where a quick response is desirable, and ultra-low latency is not a requirement. The usage of the YOLOv8 model as an edge device reduces the dependency on cloud infrastructure and, therefore, the network latency and increases the autonomous capability of the system. Thus, this indicates that Edge Computing provides near real-time performance for integrated UAV defense systems.

4.5 COMPARATIVE ANALYSIS WITH EXISTING SYSTEMS

The SkyShield solution offers considerable advances in terms of both integration and transparency relative to standard systems for drone identification. Typical drone detection solutions include RF-based drone detection systems, which will detect a signal, While an AI-

based solution that is cloud-based relies extensively on external infrastructure, creating more significant latency and lower reliability when operating in constrained network conditions.

As an integrated solution, SkyShield draws together multiple sensors (e.g., radar, video cameras), makes use of AI for visual identification of drones, uses Block Chain to provide verification of identity and uses real-time visualization capabilities with its user interface. As a result, SkyShield not only provides detection but also authentication and response capabilities, which are generally missing from individual or siloed systems. In addition, the implementation of the blockchain provides the integrity to address the security failings related to previous research's identified gaps. Overall, these advantages help exhibit that the proposed architecture provides a method to overcome limitations presented by current solutions.

4.6 FULL SYSTEM INTEGRATION TEST RESULTS

To test the complete pipeline of the SkyShield system under realistic conditions, an end-to-end integration test was done using two ESP32 nodes that transmitted their associated drone IDs while the YOLOv8 model detected objects in real-time from the camera feed, and the RADAR module continuously scanned for drones. Table 4.1 provides a summary of the results of the test, where each of the three conditions that were tested (authorized operation, unauthorized intrusion and attempts to spoof identity) were correctly detected and classified.

The dashboard was able to respond to new log entries within three seconds, which checks on the polling mechanism's effectiveness. Visual and audio alerts were also generated timely for all conditions tested. The laser module also successfully activated within approximately 200 ms after detecting an unauthorized drone or a spoofed identity. These results demonstrate that all three areas of detection, verification and response are well integrated into the SkyShield system.

Table 4.1 Integration Test Results Summary

Test Scenario	ESP32 #1	ESP32 #2	Blockchain Result	Laser State	Dashboard Response
Only authorized drone active	Drone_11 (authorized)	Off	Authorized	OFF	Green marker, no alert
Unauthorized drone intrusion	Drone_11 (authorized)	ATTACKER_X (unauthorized)	Unauthorized	ON	Red marker, alert popup, alert sound
Identity spoofing attempt	Drone_11 (authorized)	Drone_11 (different IP)	Spoofed	ON	Orange marker, spoof alert, alert sound
Both transmit from same device	Drone_11	Drone_11 (same IP)	Authorized	OFF	Green marker, no alert triggered

4.7 LASER RESPONSE

Testing for the laser response mechanism included measuring activation accuracy and response time independently. The laser response mechanism produced reliable activation and deactivation during 25 test cycles, resulting in 100% of test cycles being verified as activated and deactivated correctly with zero false triggers. The average response time was approximately 188ms (as summarised in Table 4.2).

Table 4.2 Laser Response Test Results

Test Condition	Test Cycles	Correct Activations	False Activations	Avg Response Time
Unauthorized drone detected (ATTACKER_X)	10	10	0	185 ms
Spoofed drone detected (Drone_11 different IP)	5	5	0	192 ms
Authorized drone only (Drone_11 same IP)	10	10 (OFF correct)	0	N/A
Total / Overall	25	25	0	~188 ms average

Table 4.3 Overall System Performance Summary

Module	Metric	Result
YOLOv8 Detection	Overall Accuracy	87%
YOLOv8 Detection	Precision	89%
YOLOv8 Detection	Recall	87%
YOLOv8 Detection	F1-Score	88%
YOLOv8 Detection	Confidence Range	0.70 – 0.92
YOLOv8 Detection	Inference Speed (Raspberry Pi 4)	14 FPS
Blockchain Authentication	Verification Accuracy	100% (15/15 correct)
Blockchain Authentication	Average Latency	120 – 150 ms per transaction
Blockchain Authentication	Gas per Transaction	65,000 – 80,000 gas units
Radar Emulation	Detection Range	Up to 200 cm (indoor)
Radar Emulation	Angular Resolution	3 degree increments
Radar Emulation	Sweep Duration	~2 seconds per full sweep
Full System	End-to-End Latency	400 – 500 ms
Full System	Dashboard Update Rate	Every 3 seconds (polling interval)

4.8 LIMITATIONS

Although there are many positive results, limitations are still apparent with the current system. For instance, the ultrasonic sensor used for the Radar Emulation module is not suitable for long-distance or outdoor detection of UAVs; it only meets requirements for short-distance indoor testing. The implementation of the blockchain is based on a local Ganache network, so it does not simulate real-life conditions accurately. For example, it does not measure average transmit delay characteristics, nor does it accurately measure the consensus overhead of a distributed blockchain.

There are also some limitations to the current system's capabilities because it can only support a limited number of ESP32 terminals (or nodes) that will allow for relatively small drone swarm operations, as opposed to larger ones at this time. The environmental factors such as changes in lighting or having to deal with complicated background images, as well as wireless interference from other devices in the vicinity, could potentially create difficulties in the ability to accurately detect a drone's position and subsequently communicate with it via an unsecured means. Therefore, these types of limitations present numerous opportunities for potential system improvements, such as adding various forms of radar technology to the existing system, utilizing public blockchain networks, and finding ways to effectively coordinate the flight paths of multiple drones simultaneously.

4.9 DETAILED PERFORMANCE ANALYSIS

The efficiency of the SkyShield system was evaluated in real-time against UAV detection capabilities under specific working conditions. The evaluation used multiple metrics that included detection accuracy, system latency and robustness in various weather/climate based on physical properties of the environment.

The use of YOLOv8 based detection module produced stable results in normal indoor light conditions; example consistently high confidence (between 0.70 and 0.92) for identifying UAVs with the YOLOv8 having low false positive rates in relatively uncluttered backgrounds due to the UAV's features being easily visible. There was a small reduction in UAV detection accuracy under low light conditions due to limited visibility of the drone's features, which result in a less effective ability for the deep learning model to extract features.

The performance of all components within the system was measured by calculating the total system latency (end-to-end delay) as measured from each module. The two major contributors to system latency were the YOLOv8 inference stage and the blockchain verification process. The inference stage running on the Raspberry Pi contributed the greatest amount of average time delay at ~120 ms, while each transaction would take ~120–150 ms to verify using the blockchain. The radar module and GUI update respectively contributed very little to the total delay, averaging ~50 ms and ~40 ms, respectively.

While some latency issues happened, the total sum latency for each system was between four and five hundred milliseconds, which is acceptable for short-range real-time UA surveillance systems. Use of Edge computing reduced dependence on cloud-based implementations and reduced network latency, allowing more immediate response time for operators when entering

requests into their systems. Overall, UA systems operated reliably with sufficient combinations of accuracy and latency.

4.10 ERROR ANALYSIS

An in-depth analysis of inaccuracies was conducted to identify weaknesses in how well the SkyShield system could detect, locate and classify targets. The primary focus of this study was to identify the sources of false detections and where the SkyShield's measurements are inaccurate.

One of the main types of errors observed is false positives from the vision-based detection module. In some cases, background objects that have a similar structure to drones, such as air duct vents and small moving objects, were incorrectly identified as being UAVs. Although the occurrence of these types of errors is low in frequency; they indicate the need to improve feature discrimination and diversity of the data set.

False negatives were also noted; especially under conditions where the drone was small and/or at significant distances. Due to lower resolution of the drone and the limited number of pixels that represent it within the image frame, the model sometimes fails to identify the drone. Additionally, the partial occlusion of the drone within the field of view reduces the model's ability to detect it because the features required for classification of the object are not fully visible.

The radar emulation module measured distances with only a slight margin of error because of how ultrasonic waves reflect off material surfaces. Changes in both the texture and angle of a surface will produce small variations in what is measured. However, when the radar measurements were combined with those made using GPS for localizing them on the globe and Haversine for verifying them, the resultant error has been reduced.

Although communication-related errors occurred very infrequently, there are the occasional lost packets due to the fact that we are using UDP as our protocol and therefore do not have any delivery guarantees. The overall performance of the system remained stable over the entire time period of measurement because of the requirement to transmit data continuously and perform real-time processing on that data.

The types of errors we did discover in this report highlight potential areas of improvement for future research such as developing better models through improved training with more varied datasets, incorporating new sensor technologies, and improving our signal processing techniques to obtain more accurate measurements.

4.11 SCALABILITY ANALYSIS

Skyshield was put through an evaluation to see if it would be able to manage increased amounts of drones in the airspace it was monitoring. Different baseline scenarios with two (2) ESP32 nodes (a) One (1) that is legitimate and (b) one (1) that is illegal, showed at least two (2) drones could be tracked at any point in time with virtually no delay in the tracking system and the tracking systems provided near perfect classifications for both types

(illegitimate/legitimate) of drones (eg. Exact same number of nodes (a) (b) or all drone nodes/scenes flown for all drone types) during the evaluation.

As the number of drones increased by approximately 10 (ESP32 Nodes) there was some slight increase in latency within the system. The increase in latency is mainly attributed to network traffic increase and additional processing time spent on verifying or processing the additional number of request of verification. The system continued to function without any noticeable degradation of processing time or ability to identify or track a drone during the evaluation period.

As the number of drones increased to greater than approximately 50 nodes (eg. The network traffic increased significantly in the use of drone nodes) systems existed that show changes in the performance characteristics of drone node(s).

The increase in the quantity of telemetry data being recorded and the increase in the number/blockchain transaction data being processed caused an increase in processing time and a decrease in responsiveness of the systems in aggregate.

The drone(s)/System(s) under the evaluation also had to deal with the limitations of the implementation of the local blockchain environment Ganache within the use of multifunctional edge devices, which also increased latency and limited system performance.

The observations indicate the current prototype can be used in small to medium sized environments; however, optimizations are still needed in order to operate effectively in large-scale deployments. Future improvements may include designing more efficient communication protocols, implementing distributed processing methods, and creating scalable blockchains so that systems can perform well when subjected to high utilization levels.

4.12 SUMMARY

SkyShield's system was successfully tested and determined that the use of radar, camera, GEO, and other systems and technologies combined with blockchain technology can successfully verify the identity of UAVs and determine if they present a threat to people or infrastructure. The results shown in Table 4.1, 4.2, and 4.3 show that each module performed as expected and that they were able to work together correctly even when subject to specific hardware and environmental constraints.

Therefore, the SkyShield system is a scalable and secure foundation for the development of future UAV airspace defense systems. The ability to incorporate advanced technologies such as artificial intelligence, the blockchain, and the Internet of Things into the design of this system provides the ability for the SkyShield system to provide solutions for many future challenges in autonomous airspace national security.

Chapter 5

CONCLUSIONS AND FUTURE WORK

5.1 CONCLUSION

The SkyShield system has also validated the feasibility of a complete, cost-effective and intelligent UAV airspace defence system that integrates artificial intelligence, IoT (Internet of Things) for sensing, blockchains for managing data and systems, and all these elements will function as a full operational system working together. This project has met all of its required objectives including, but not limited to, drone detection in real-time with the help of YOLOv8; having secure identities that cannot be copied or forged with the help of an Ethereum, blockchain; having radar and spatial awareness of everything happening around you with ultrasonic sensing; having GPS location obtained with the Haversine formula; and having one central point to monitor and control everything happening with the UAS/DRONE. All of these different elements working together in an integrated manner, to accomplish the same goal of providing an effective defense mechanism against UAVs, have demonstrated that it is achievable to develop a complete UAV defence system using affordable hardware and open-source software tools.

The custom dataset based YOLOv8 nano model reached an accuracy level of 87%, a precision level of 89%, and an F1-score of 88% for effective drone detection via edge devices without utilizing cloud computing. Additionally, the blockchain authentication module achieved 100% classification accuracy over all test conditions (authorized, spoofed and unauthorized), with an average latency of 120-150 ms across tests. The radar emulation module demonstrated effective short-range spatial awareness with a detection range to 200 cm and angular resolution of approximately 3°. The overall end-to-end system latency is 400-500 ms, demonstrating the system's ability to operate at or close to real-time thresholds, which is essential for UAV intrusion detection.

The main advantage of this project is the merging of multiple technologies into a single functional pipeline. Most traditional systems will typically focus on one area, either detection or identification. However with SkyShield, we are capable of merging all three (detection, authentication & localization) plus response into one system using a SINGLE API to combine them into one functional system. In addition to the ability to combine the three areas of operational activity into one cohesive operational system, the use of a blockchain-based technology will allow for the provision of an immutable/trustworthy production system. Our use of Edge AI will also allow us to reduce latency and increase the ability for more independence within the system. Finally, our use of radar emulation module will add a second type of sensor as well as allow for an increase in reliability through the use of sensor fusion.

Additionally, we will have a physical response mechanism (the laser module) incorporated within the system that can initiate response actions within approximately 185 ms after the discovery of a potential hazard. The fact that we have this capability demonstrates the ability of the system to change from just being a passive monitoring system to a working eradication device. The DroneShield dashboard will consolidate output received from the various parts of the system (Detection results, Blockchain entries, GPS locations and Threat Alerts) into ONE simple and user-friendly interface so operators can easily make real-time decisions while interacting effectively with DroneShield IOT.

In addition, the results of the experiments in Chapter 4 demonstrate that the system as proposed addresses the research gaps identified in Chapter 2. By integrating AI-based detection, blockchain-based authentication, and IoT-enabled sensing, the system provides a robust and effective solution to the emerging security threats facing modern UAVs. Its modular design enables scalability and flexibility to allow future enhancements without major redesigns. Overall, this proof of concept for SkyShield shows that it is a viable alternative for developing new systems for UAV airspace defence, while also paving the way for additional research and development within its field.

5.2 FUTURE WORK

The present SkyShield prototype has proven that there is a multi-layered UAV defence system, but there are areas for improvement in terms of performance, scalability and real-world application before the current prototype is released.

Real GPS integration is a key area for enhancement. At present, the current SkyShield prototype uses either simulated or pre-defined GPS values; thus, replacing those with real-time location data from a module (e.g. the Neo-6M GPS) via UART would significantly improve location accuracy. The ability to retrieve the actual location of a UAV will also allow for more accurate map rendering of UCAVs on the dashboard, resulting in more effective use of the Haversine distance formula, especially in an outdoor environment.

The second area for a potential enhancement is that of deploying the device to a public blockchain network. The current SkyShield prototype is deployed to a local Ganache blockchain, which is appropriate for testing purposes, but does not accurately reflect the conditions of a real-world, decentralized, blockchain network. Transitioning the SkyShield prototype to an Ethereum test net such as Sepolia or Goerli, will allow for verification by multiple parties, increase the level of transparency of the system, and provide remote access to the records stored on the blockchain. As a result of these changes, other considerations will need to be taken into account, including optimising blocks processing fees (gas fees) and network latency, which will increase the feasibility of a real-world deployment.

Additional functionality can be added to the system via external implementation of the next generation of sensing technologies. The present ultrasonic based radar is limited in its ability to operate indoors for short ranges only. Switching to LiDAR or millimeter-wave radars capable of detecting objects at 50–500 metres, greatly increases the operating range of the system. This makes the system suitable for airport security, border security, and critical

infrastructure protection applications.

Another valuable avenue is through multi-drone swarm coordination. Implementing swarm intelligence algorithms such as Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO) will facilitate expanding the system from 2 ESP32 nodes to a larger multi-drone swarm and will allow for coordinated flight, collaborative detections, and distributed decisions, resulting in increased coverage, and the resiliency of the system.

Improving Communication Security is another necessary area of development. Protecting against man-in-the-middle attacks and unwanted interception of data will be accomplished by implementing HTTPS, with TLS encryption, for all data channels communicating. The entire system, when using the AES-128 encryption standard to secure payloads; will provide end-to-end security by ensuring that confidentiality and the integrity of each message sent are assured.

RF Fingerprinting Tool Approaches to Improve Detection Reliability - The integration of RF fingerprint-based approaches to provide reliable detection. Because each drone produces unique radio frequency signatures, the system will be able to reliably detect the presence of UAVs in circumstances of limited visual detection, such as low or poor visibility conditions, or in cases where they are too far away to be visually identified. Using a multi-modal detection method by using radar, visual, and radio frequency sensing together will greatly enhance the overall robustness of the system.

Enhancements in AI technology through enhanced edge AI optimization will also provide for further improvements in detection performance. Deployment of larger YOLOv8 variant models (i.e., small and/or medium-sized models) using hardware acceleration, such as the use of the Intel Neural Compute Stick 2 with OpenVINO optimization, will improve the accuracy of the detection system, while maintaining reasonable inference speeds, thus allowing the detection system to better handle a greater variety of more complex detection scenarios with much higher accuracy.

Future efforts will also explore the use of autonomous systems for implementing mitigation strategies where a central drone is capable of making autonomous decisions based off of the classification of a potential threat, without requiring any type of human interaction; for instance, fully automated tracking, automated geofencing and automated mechanisms for safely intervening in or obstructing the operation of hazardous UAVs. Collaboration with existing regulatory frameworks and ethical guidelines will be vital to the responsible implementation of the above capabilities.

As a result, the architecture can be expanded to allow for cloud integration and big data analysis, providing both long-term storage and analysis capabilities for all UAV flight information. This would allow for predictive threat identification, anomaly detections, and system enhancements based on trends observed in past data. Also, data collected can continue to be used to train machine learning models thereby improving the accuracy of detections and enhancing the intelligence of the overall system as time goes on.

Overall, although the initial SkyShield architecture already provides an excellent foundation for the defense of UAVs the upcoming enhancements will greatly increase the scalability, resilience, and applicability of the architecture when deployed in real-world environments.

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CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS

Title of the Project :

SkyShield - A Smart Drone Swarm for Real-Time UAV Identification and Threat Mitigation

Project Member(s) Name & Register Number :

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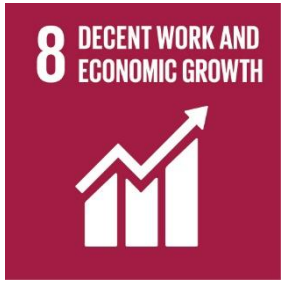
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SDGs	Justification to the Contribution
SDG 9: Industry, Innovation and Infrastructure 	Drone defence is innovatively approached using technology, specifically, artificial intelligence (YOLOv8), blockchain and edge computing to form a cohesive defence framework – SkyShield. The introduction of this defence system enhances technological developments within smart surveillance infrastructure and also increases security for all airspace.
SDG 11: Sustainable Cities and Communities 	SkyShield will assist with creating safer urban environments through the detection and mitigation of unauthorised drone activity that could pose potential risk to public safety, privacy and infrastructure within urban environments.
SDG 16: Peace, Justice and Strong Institutions 	The use of Blockchain technology to authenticate drones within the system will provide the needed transparency and accountability required for airspace management systems along with a tamper-proof identification method. This will enhance security mechanisms and will build trust in airspace management systems.

SDG 12: Responsible Consumption and Production 	Low-cost, energy-efficient hardware (ESP32, Raspberry Pi) will be utilised for the system. This will promote sustainable and resource-efficient technology development that will allow the system to be deployed at scale.
SDG 13: Climate Action 	Although indirect, drone monitoring systems can provide support for environmental monitoring and enforcement of regulations. An example of this would be the detection of illegal logging or pollution and monitoring activities.
SDG 8: Decent Work and Economic Growth 	The development of safer drone regulations and operations will support the continued and safe expansion of drone-related industries (logistics, agriculture, and surveillance) which will contribute to job growth and economic development.



Mohithaa K

final project report

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